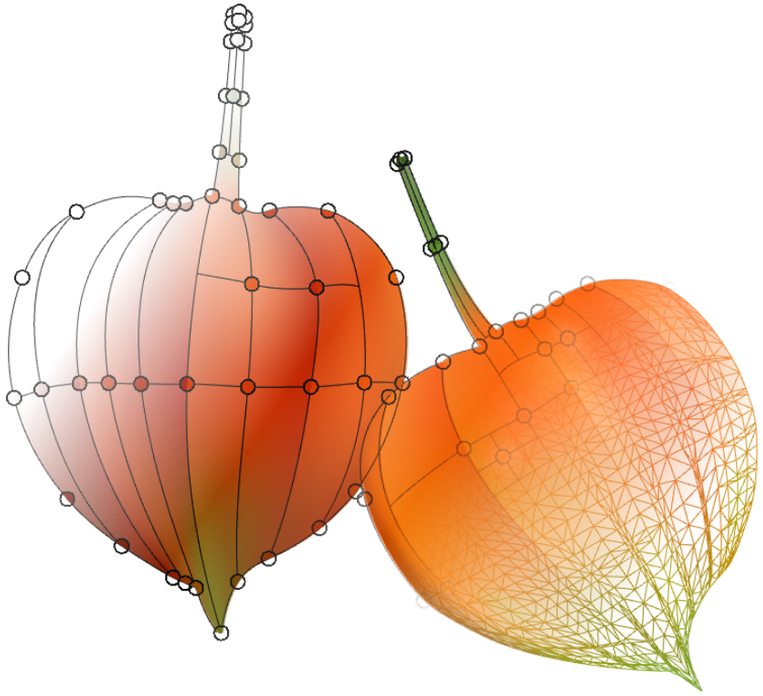
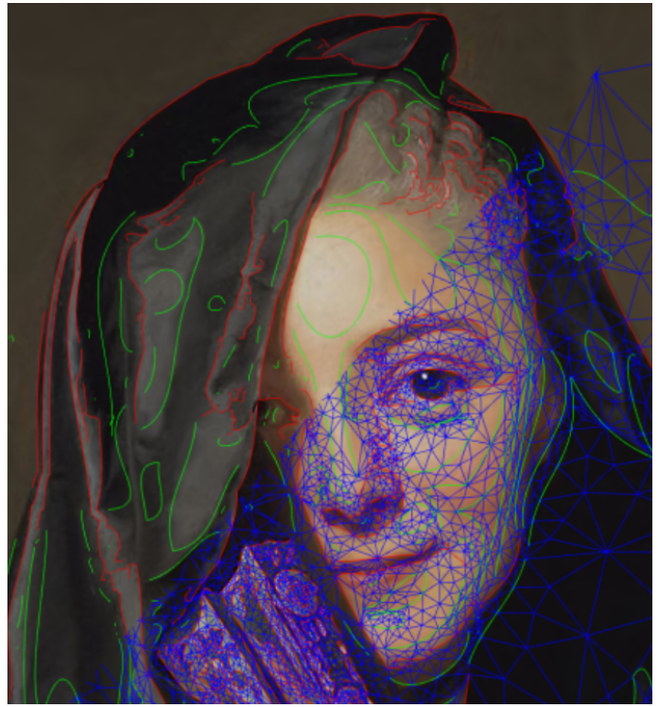
Scientific Visualization and Computer Graphics

Ruslan Khasanov





Jiří Kosinka

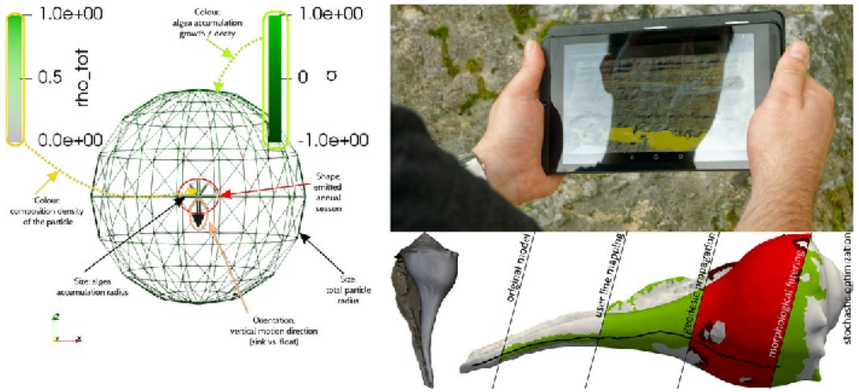
Introduction to Computing Science
10 September 2026

www.cs.rug.nl/svcg/People/JiriKosinka

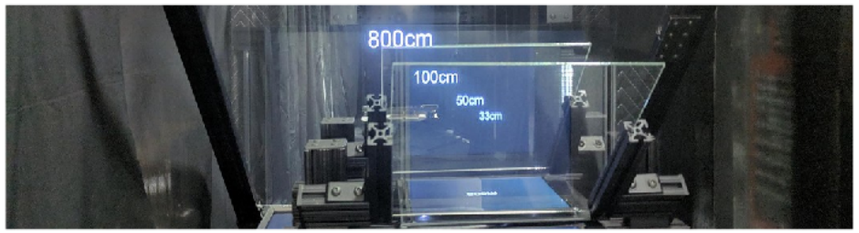
university of groningen

J.Kosinka@rug.nl

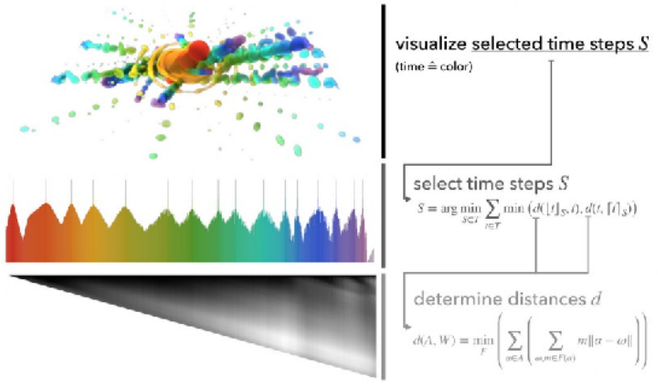
Scientific Visualization and Computer Graphics



Visualization- and Interaction Design



Perceptually optimized display systems



Optimization of scientific large data visualization

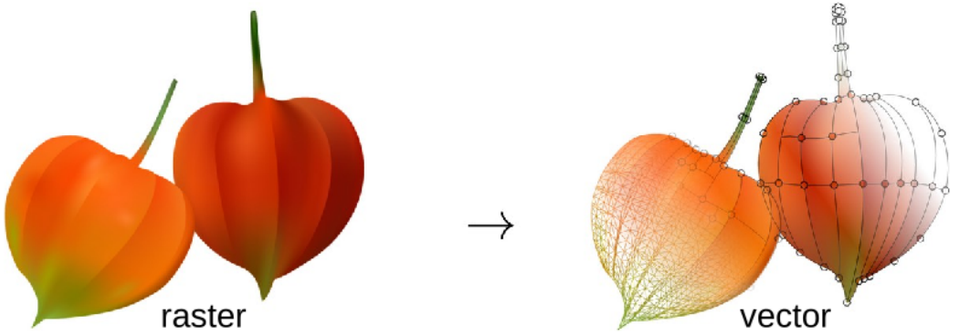


Image vectorization (fully scalable, sparse, easy to edit)

General topics and interests

- Shape and image modelling and processing
- Big data and high-performance visualization
- Optimization and machine learning
- Visual perception
- VR/AR display technology, eye tracking
- Data-driven method & system configuration
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- Medical data, astronomy, …

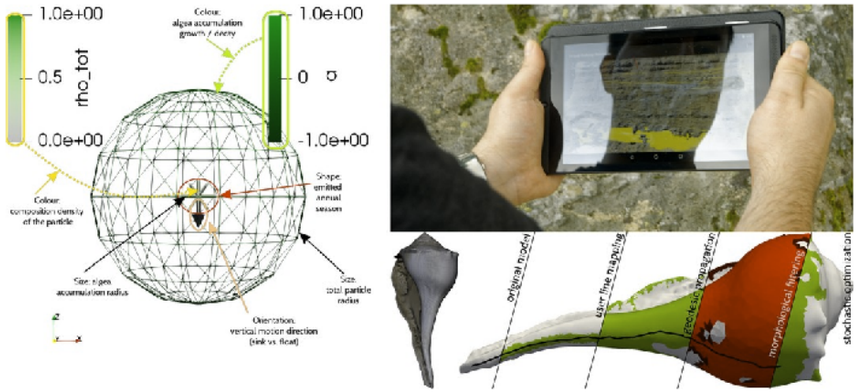
Steffen Frey

Jiří Kosinka

Jos Roerdink

Alex Telea

Scientific Visualization and Computer Graphics



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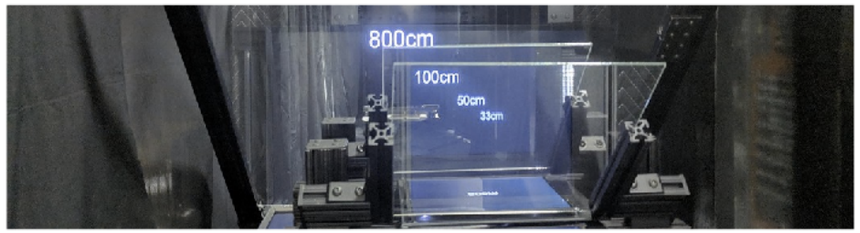

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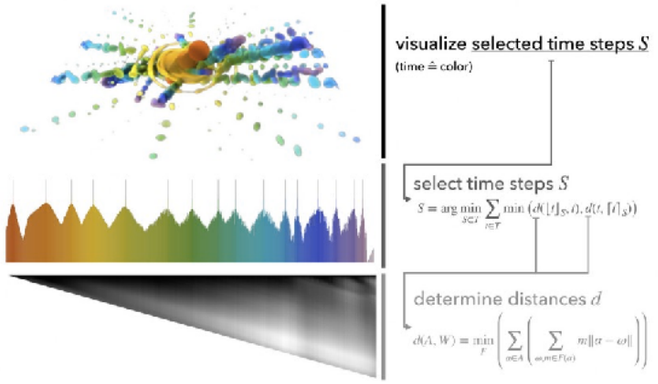

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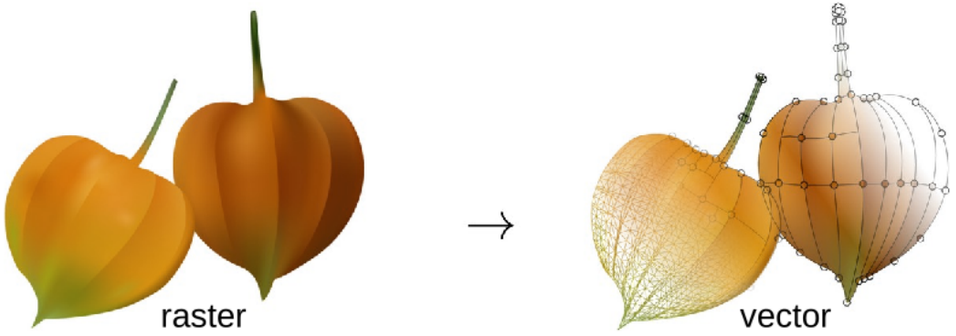

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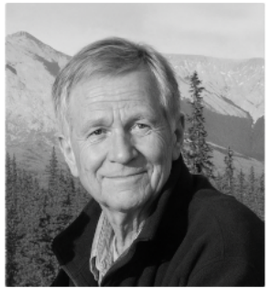

Jiří Kosinka

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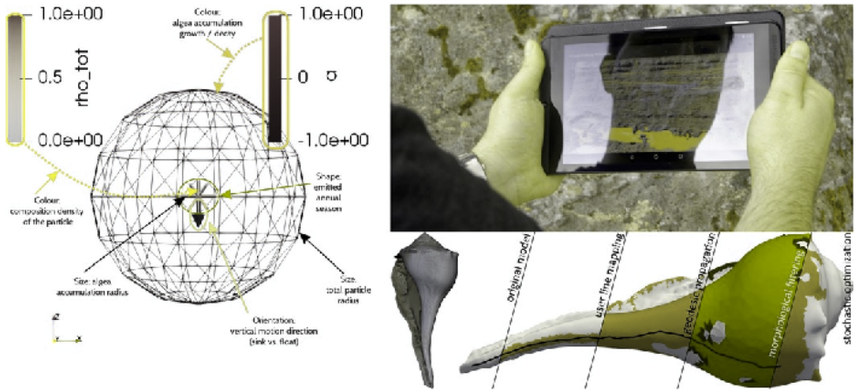
Alex Telea

Scientific Visualization and Computer Graphics



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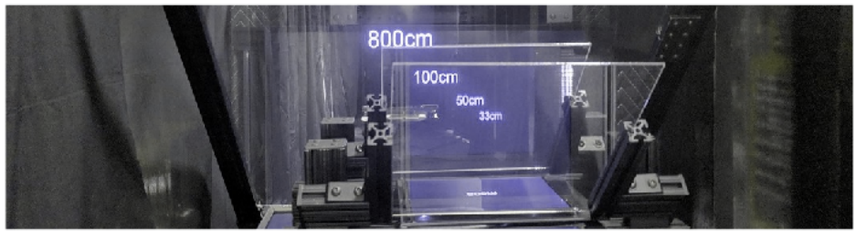

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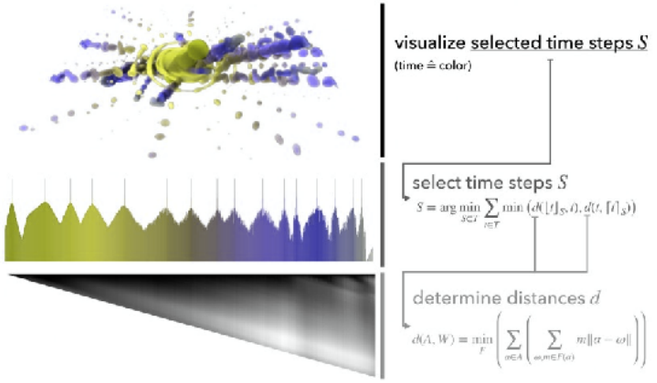
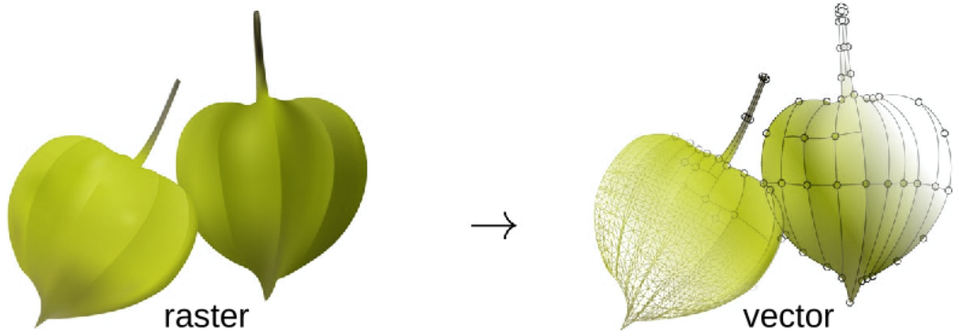


Optimization of scientific large data visualization



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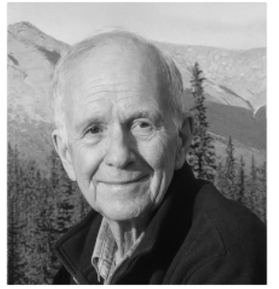


Jiří Kosinka

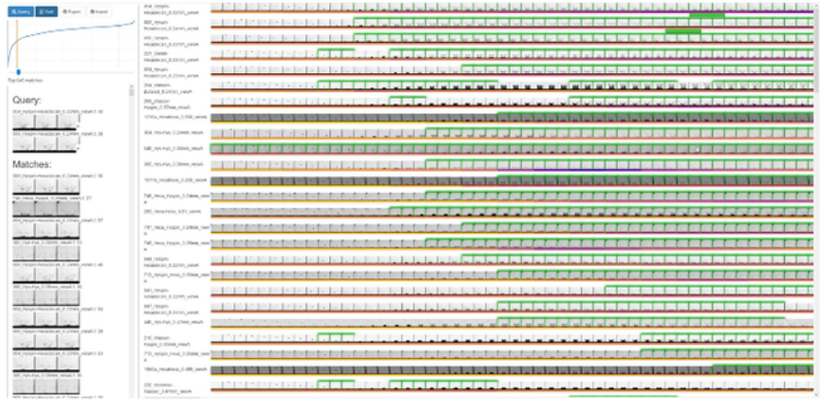
Jos Roerdink

Alex Telea

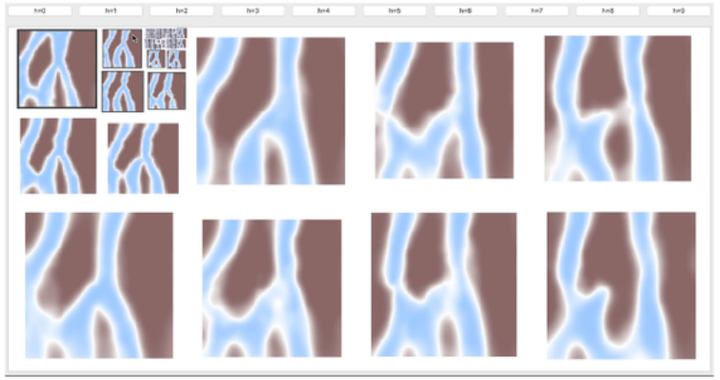
Scientific Visualization and Computer Graphics

Steffen Frey

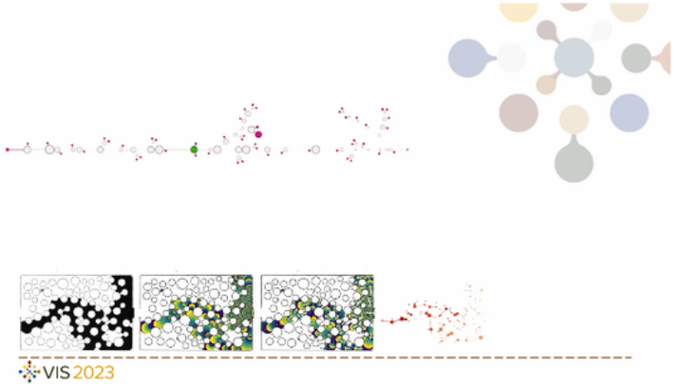
Large Data Visualization



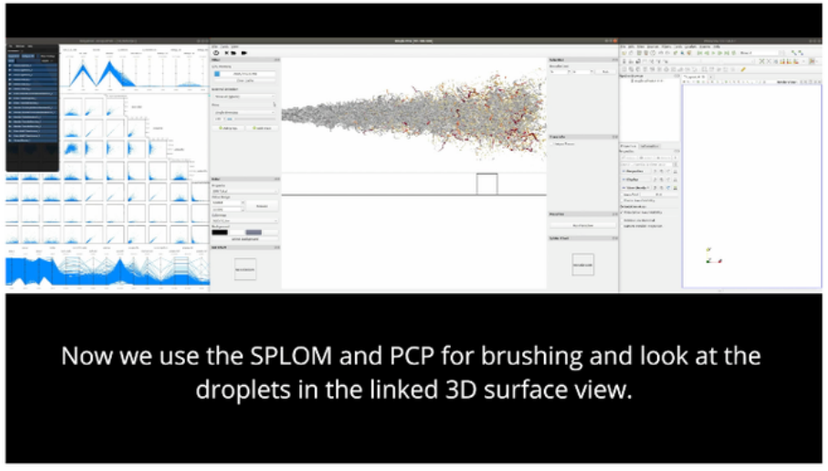
Visual analysis of droplet dynamics in large-scale multiphase spray simulations



Grid Layouts for Level-of-Detail Exploration of Large Data Collections



Porous Media Analysis



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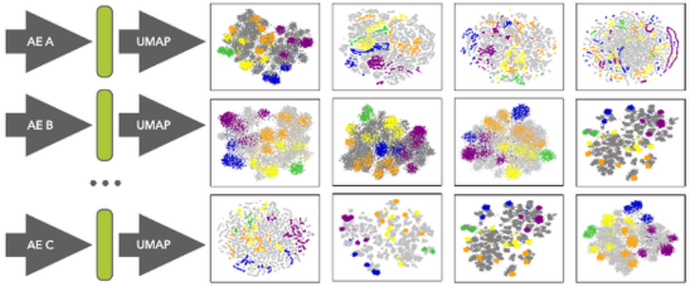
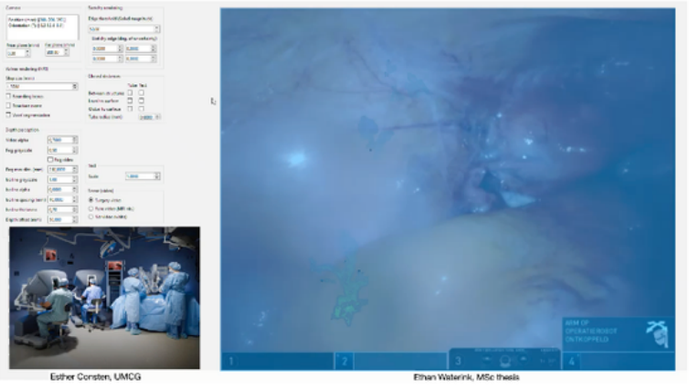


Dimensionality Reduction

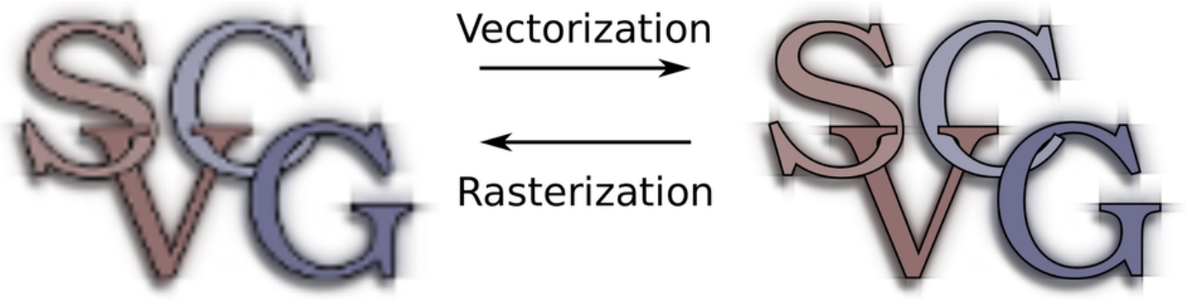


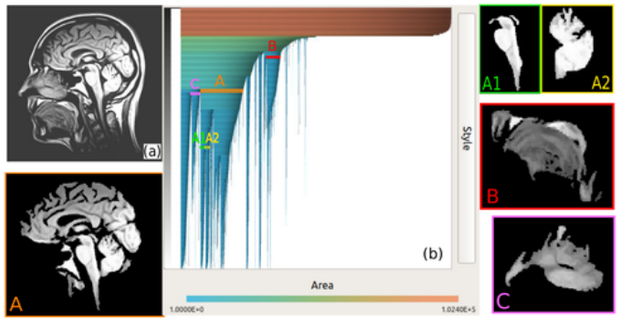
Medical Visualization

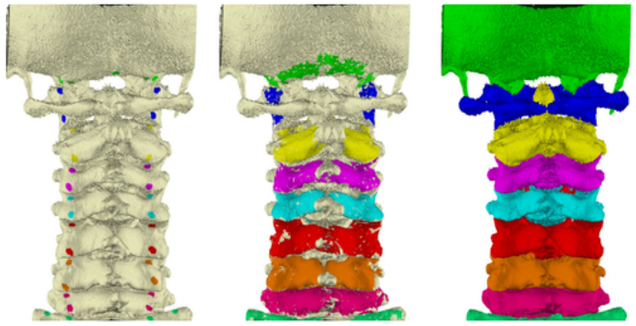
ML-driven Data Analysis | **Visualization of Large Data Collections** | **Interdisciplinary Collaborations**

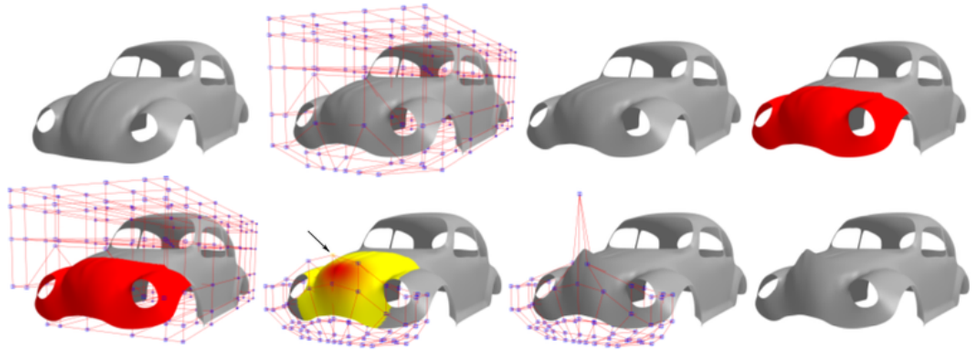
Scientific Visualization and Computer Graphics

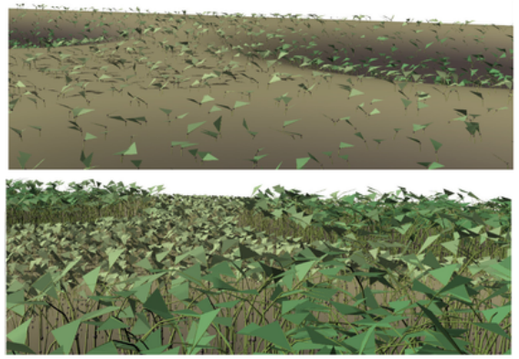
Jiří Kosinka
Shape & image modeling & processing

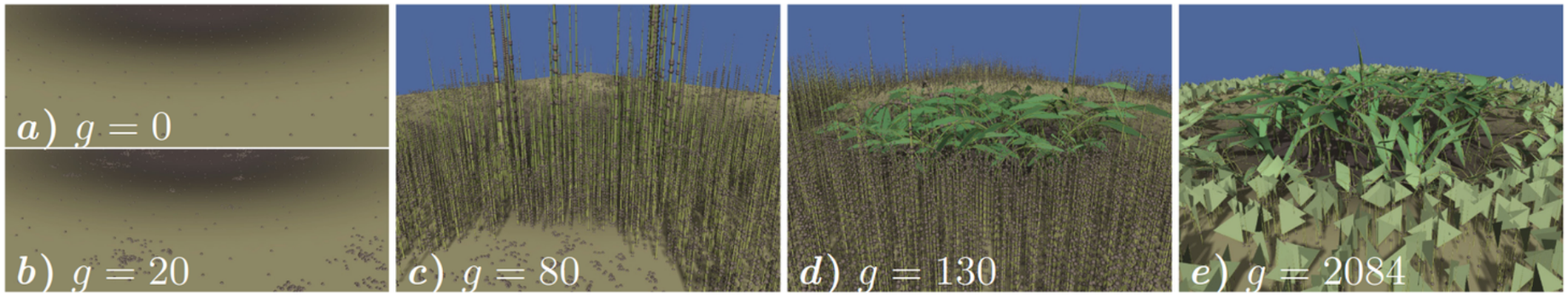




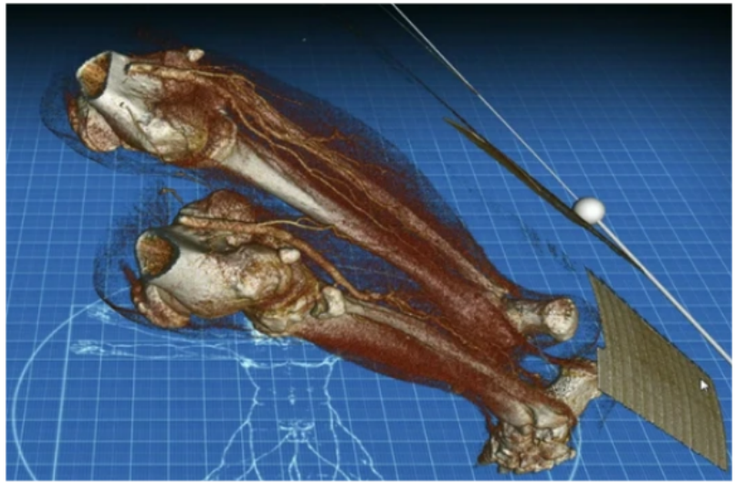


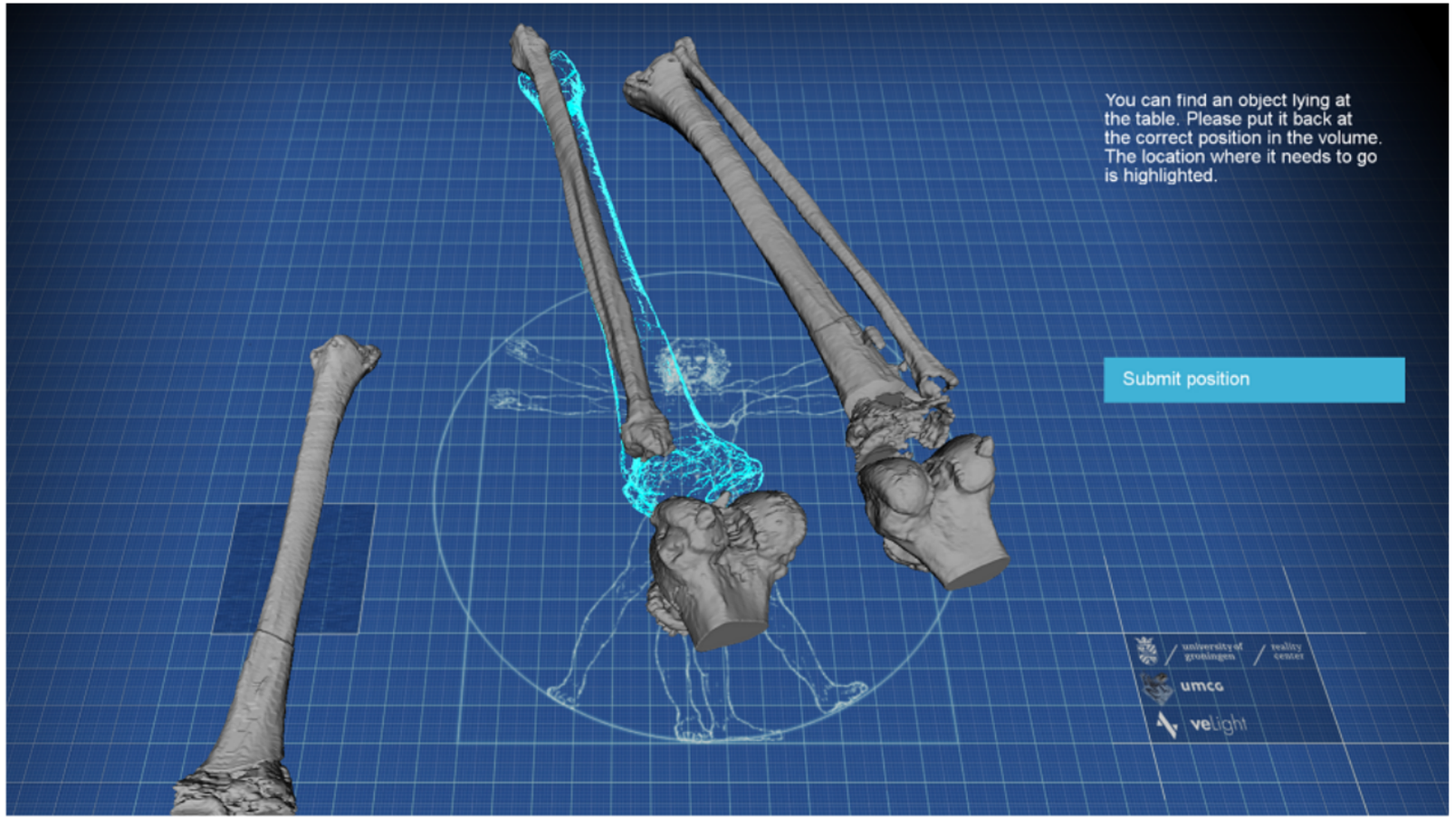






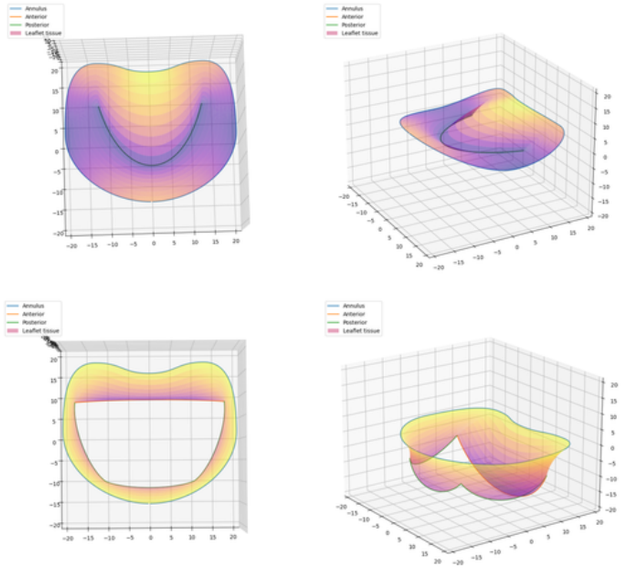
Scientific Visualization and Computer Graphics



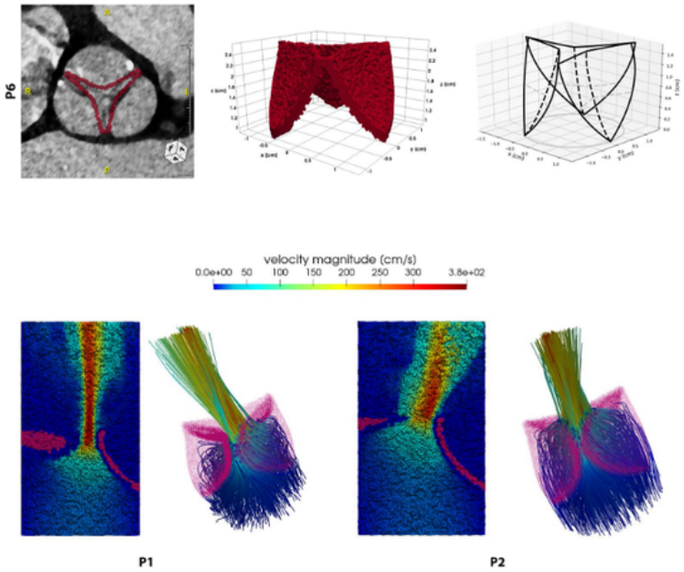


VeLight: anatomy education

Scientific Visualization and Computer Graphics



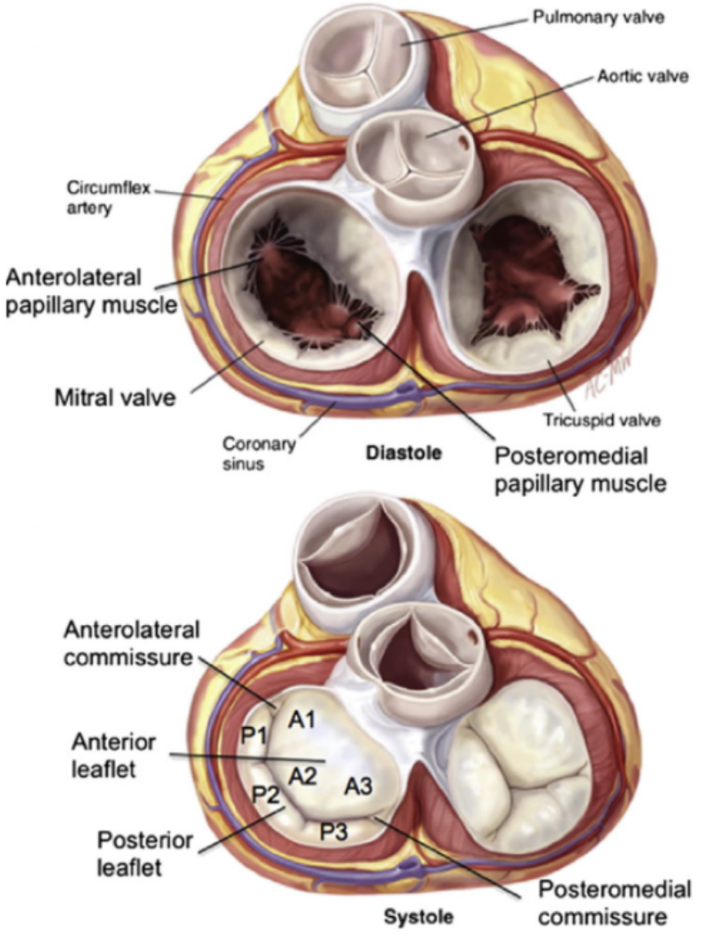
Bob vdV.



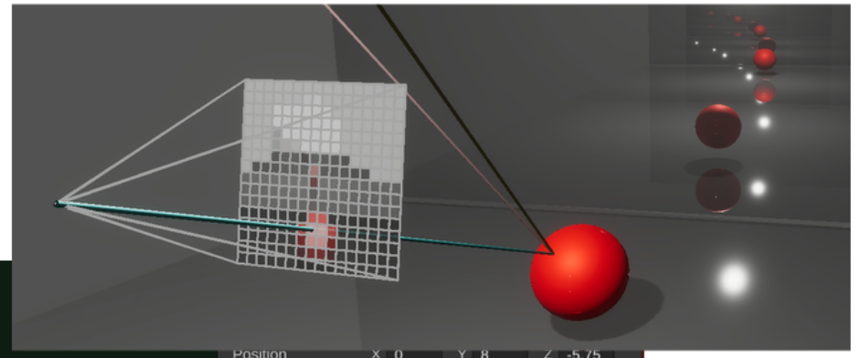
Georgia P., Tanja dV., Cristobal B. (maths)

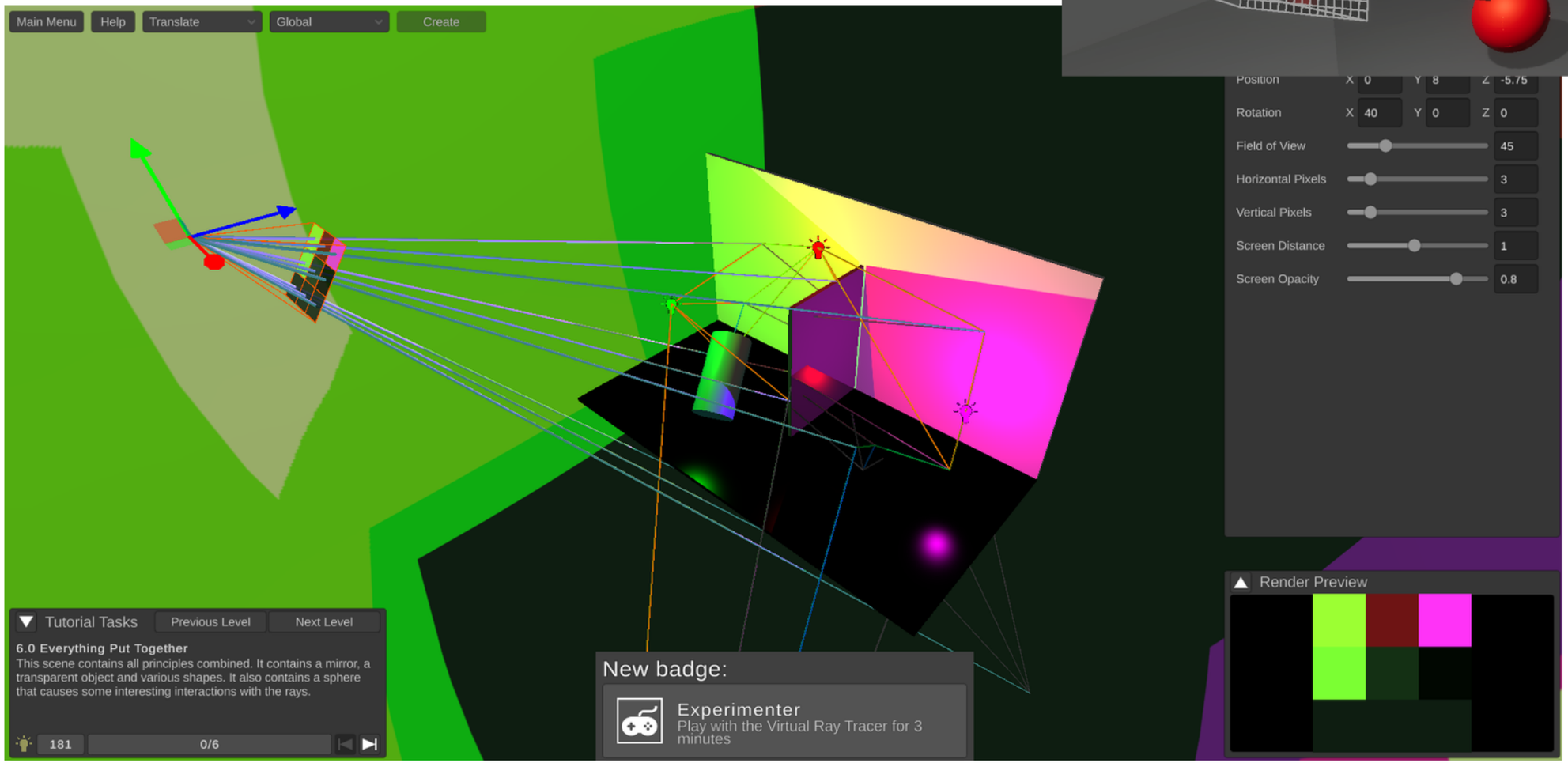
Mitral and aortic valve incorporation in real heart geometries, so that we can run blood flow simulations.

Potential extension: estimate valve shape parameters from velocity measurements.



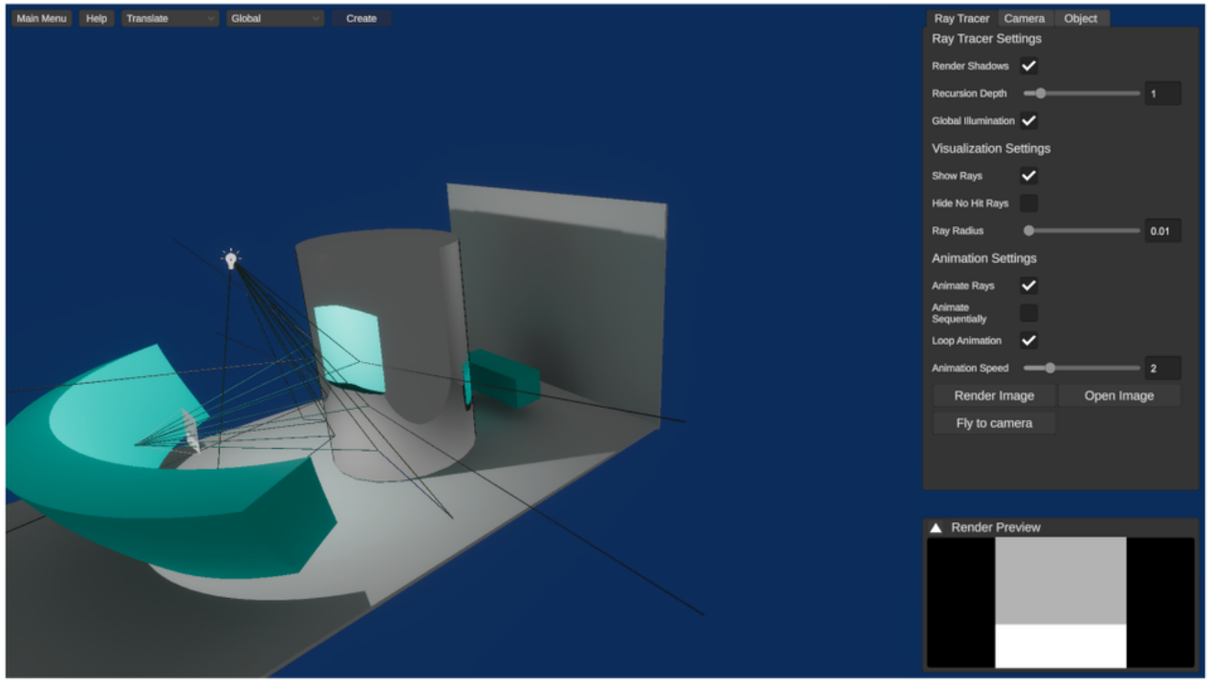
Scientific Visualization and Computer Graphics





Virtual Ray Tracer: https://www.cs.rug.nl/svcg/VRT

Scientific Visualization and Computer Graphics



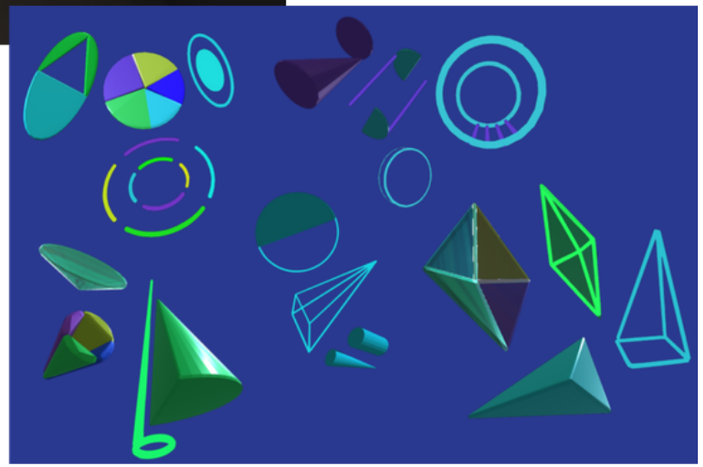
[Virtual Ray Tracer](#) - Anamorphic sculptures

Can we correct the shading?

Perhaps by recoloring the (un)deformed object?

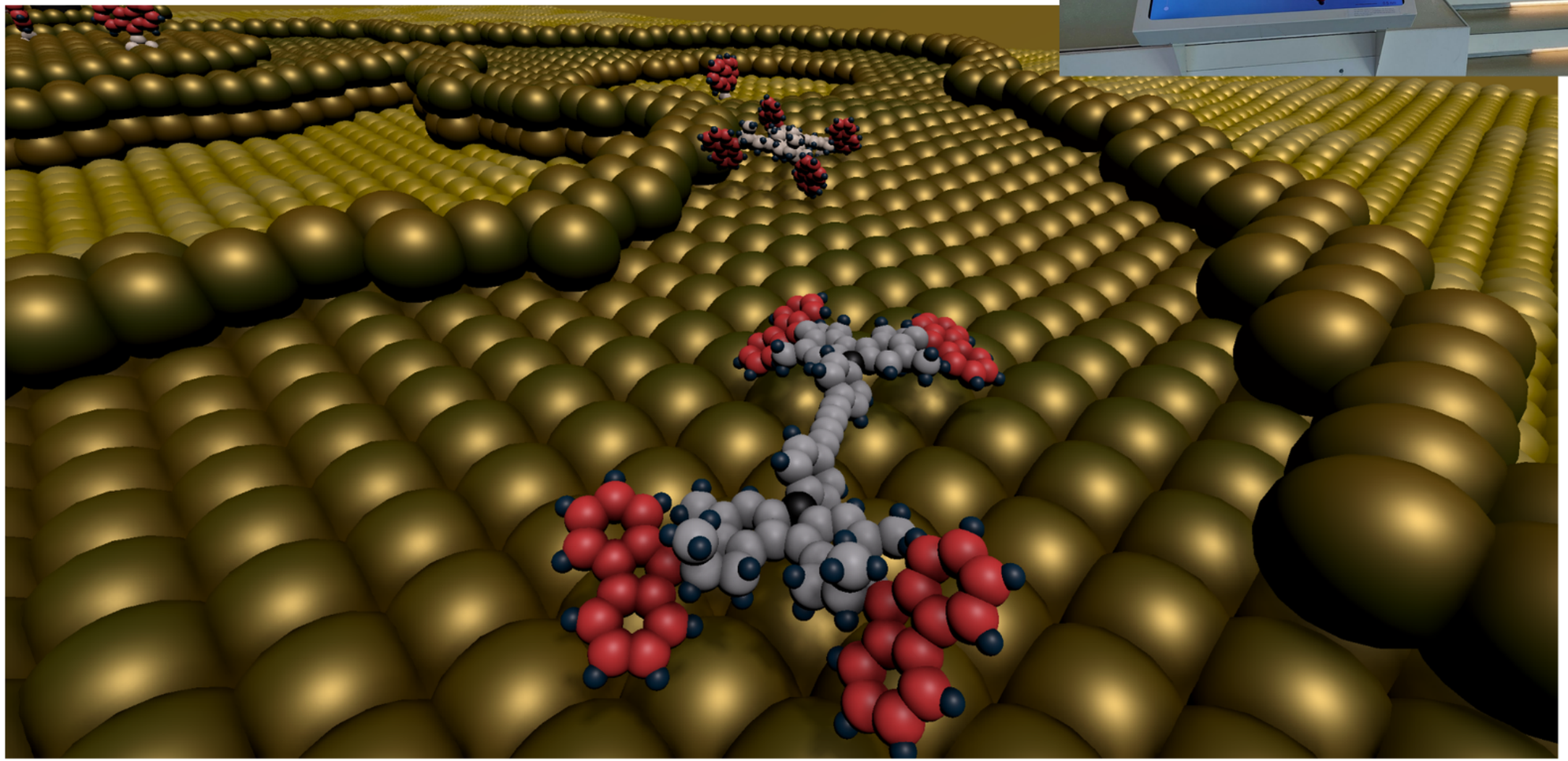
Chingiz D.-K.

Vlad C.



How precise is drawing in Virtual Reality?

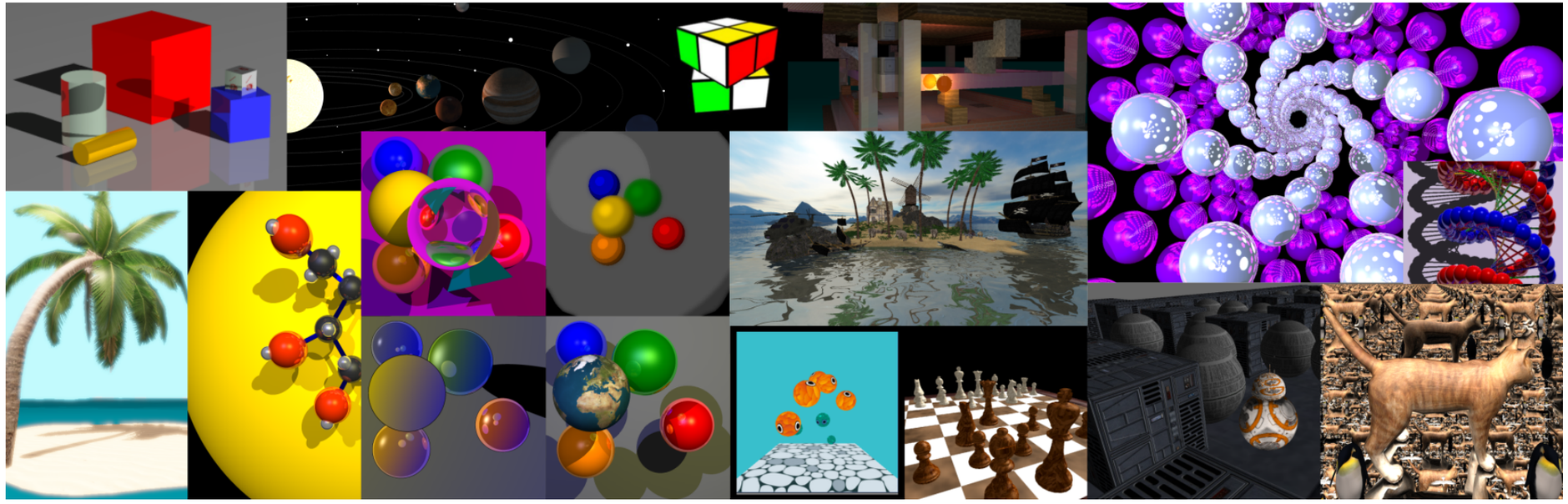
Scientific Visualization and Computer Graphics



Nanocar Grand Prix: https://molecular-car-project.github.io/WebBuild/Combined/

Teaching

- Courses we teach
  - Algorithms and Data Structures in C (for CS)
  - Research Skills in Computing Science
  - **Introduction to Computer Graphics and Visualization**
  - **Computer Graphics**
  - **Advanced Computer Graphics**
  - **Scientific Visualization**



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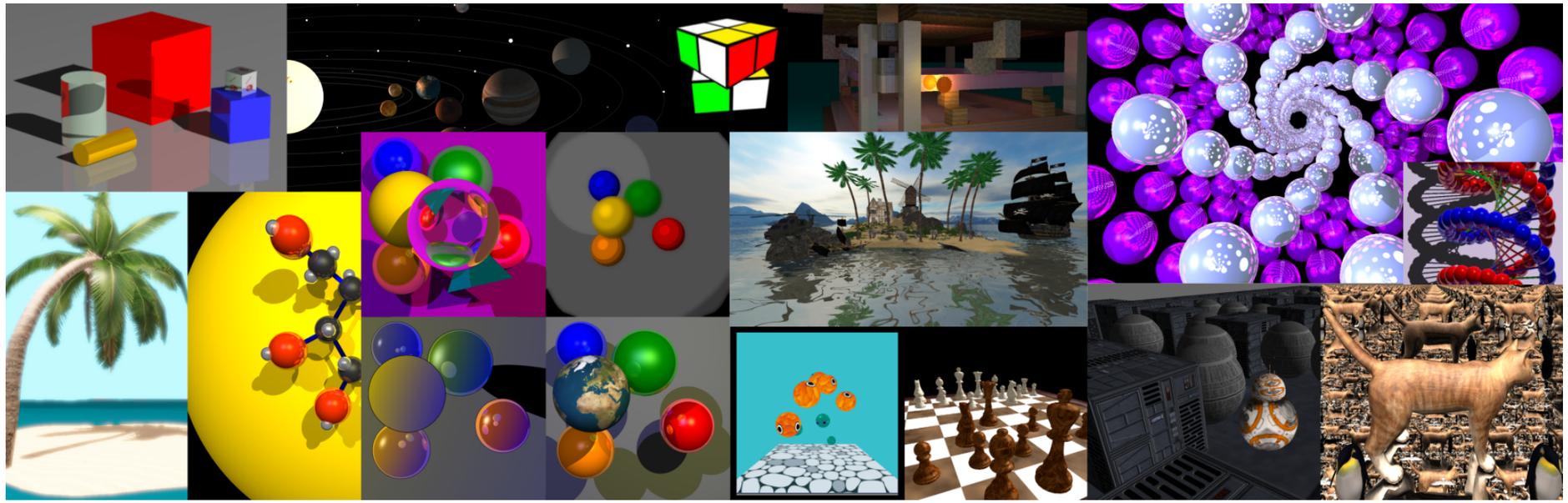
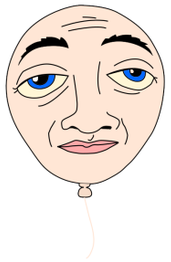
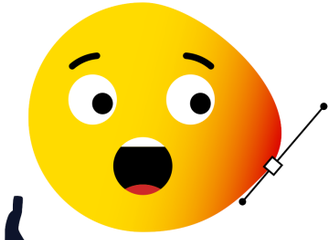
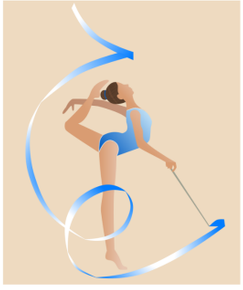
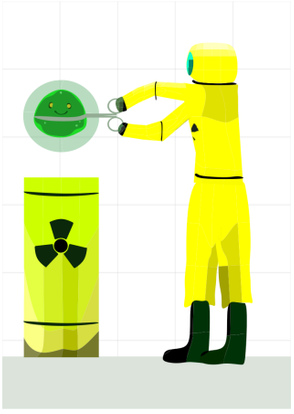
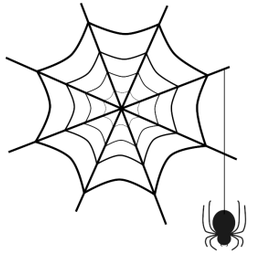


Vector Graphics Tutorial – Student Results





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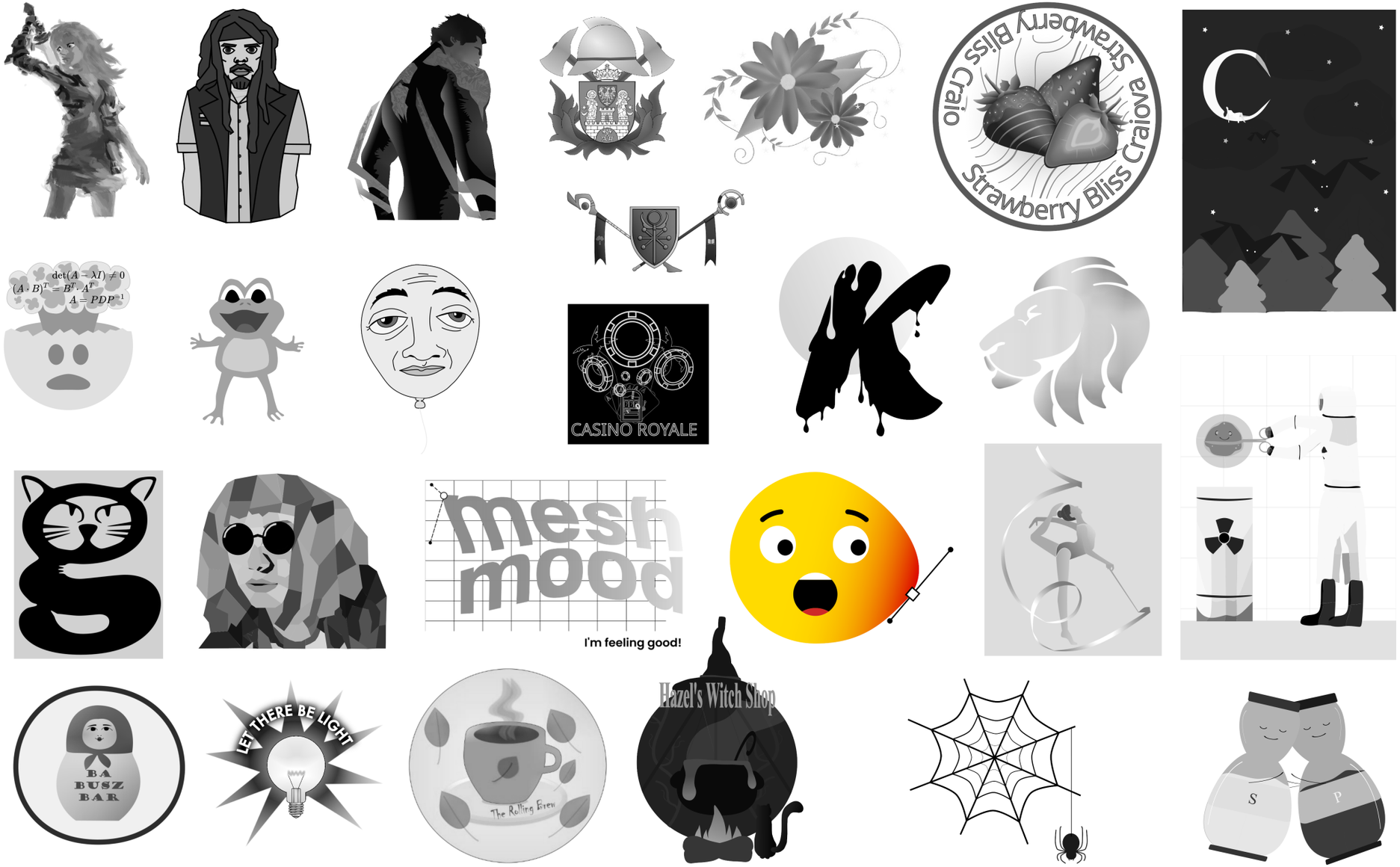


Vector Graphics Tutorial – Student Results

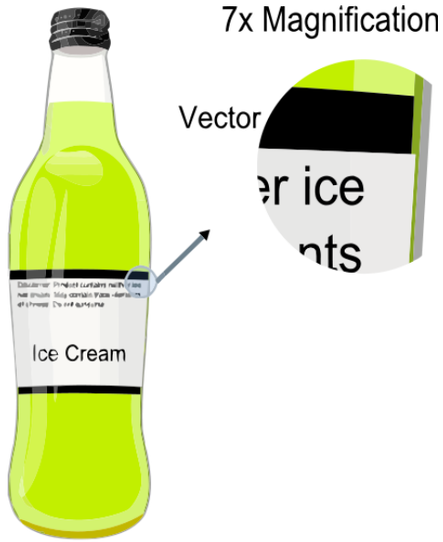
Raster vs. Vector Graphics

Raster vs. Vector Graphics

Digital Images

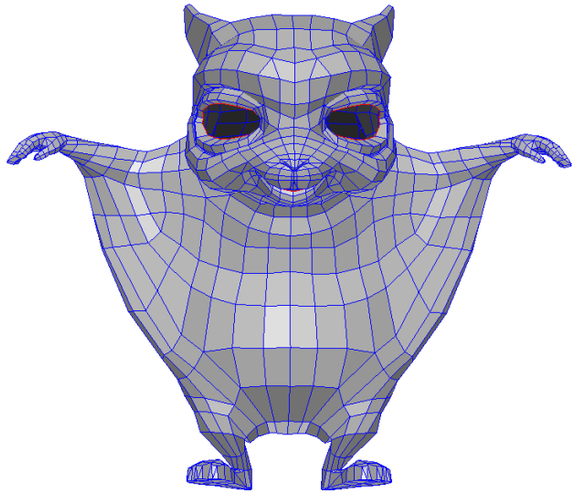
- Raster (bitmap) images
  - Array of picture elements, pixels
  - Rely on rasters
  - Typical example: .jpg, .bmp
  - Fixed resolution
  - Need to be resampled for display

- Vector images
  - Described by ‘vectors’
  - Rely on (smooth) geometry primitives
  - Typical example: clip art
  - Resolution independent
  - Need to be rasterized for display

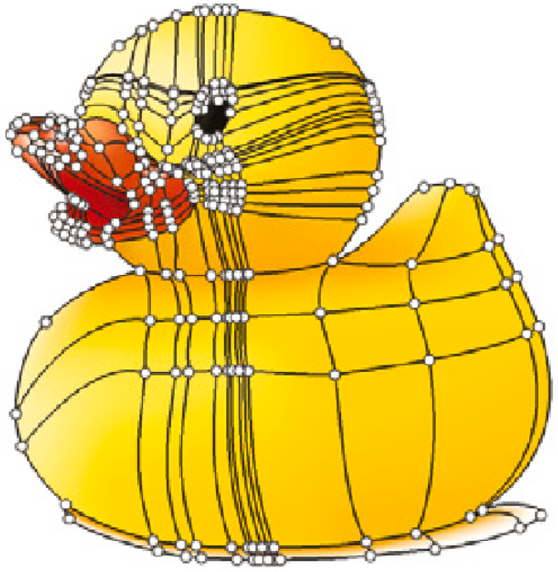


Vector Graphics

- Geometry modeling & fitting
  - is often hard
  - 3D: x, y, z



- Vector graphics & vectorization
  - is easy
  - 2D: x, y



Vector Graphics

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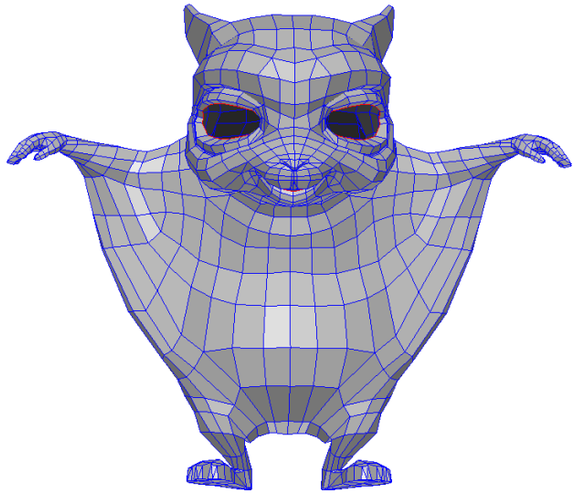
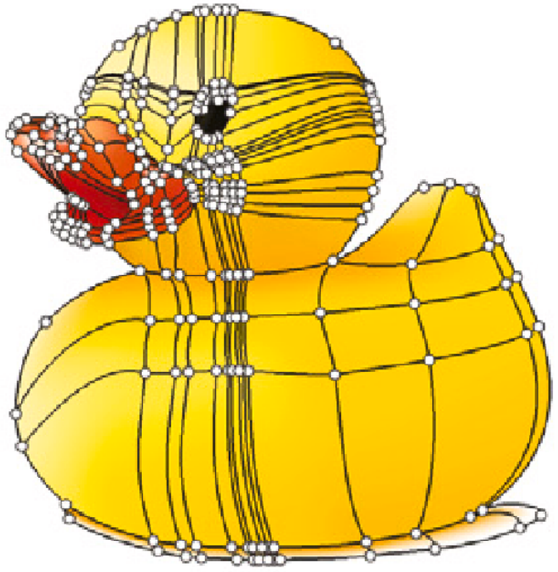
  - is often hard
  - 3D: x, y, z

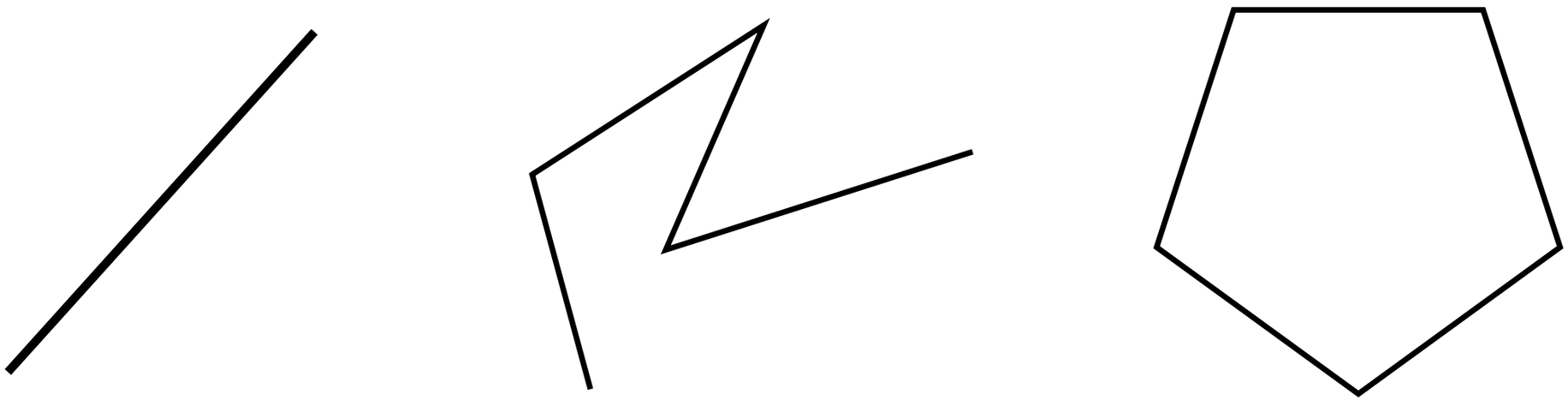


- Vector graphics & vectorization
  - is ~~easy~~ even harder
  - 5D: x, y & r, g, b



Vector Graphics

- Vector graphics primitives
  - **Lines, polylines, polygons**
  - Circles, ellipses
  - Bézier curves (cubics), bezigons
  - Diffusion and shading curves
  - Color gradients



Vector Graphics

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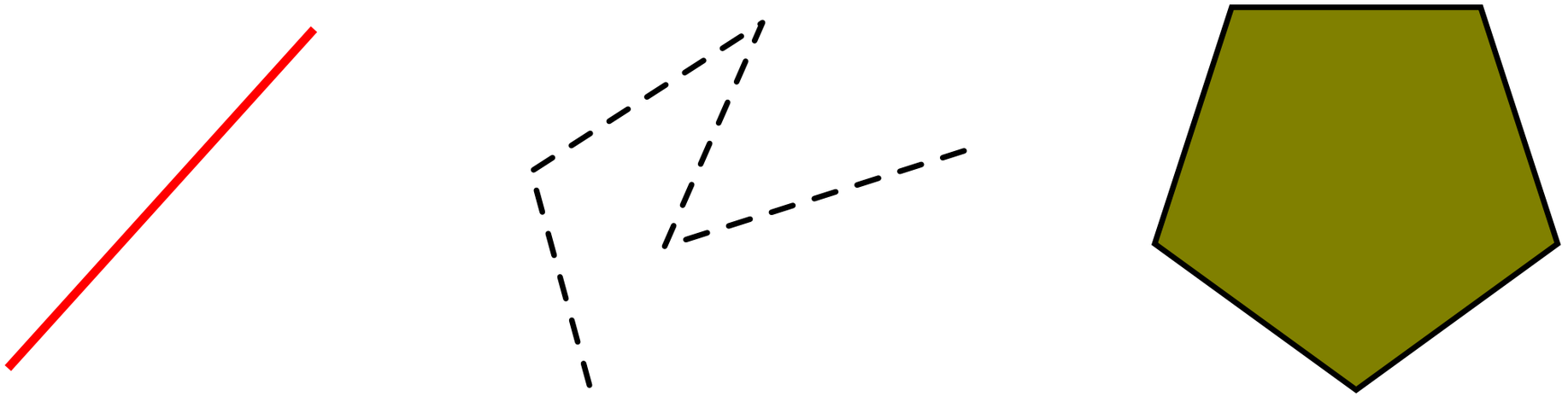
  - Circles, ellipses
  - Bézier curves (cubics), bezigons
  - Diffusion and shading curves
  - Color gradients



Vector Graphics

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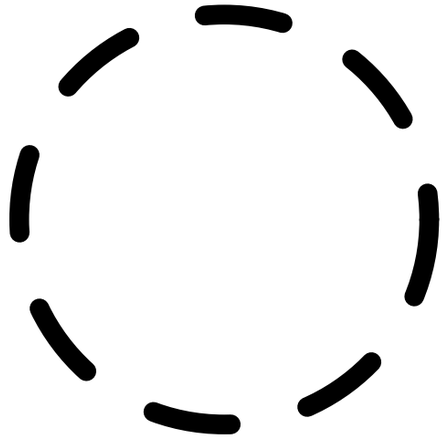

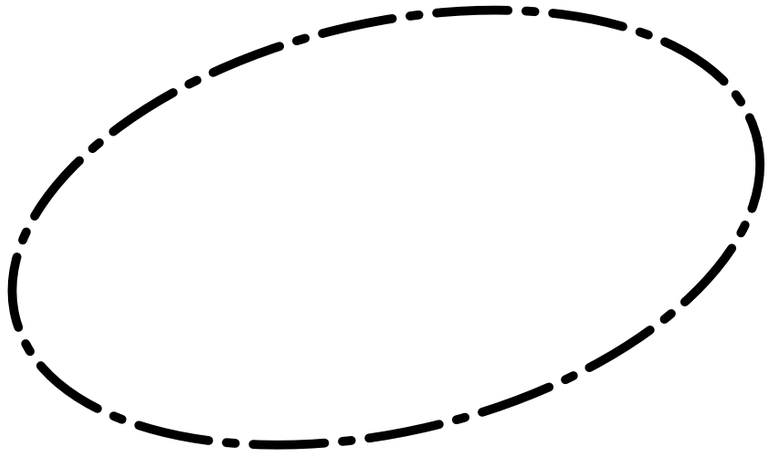

Vector Graphics

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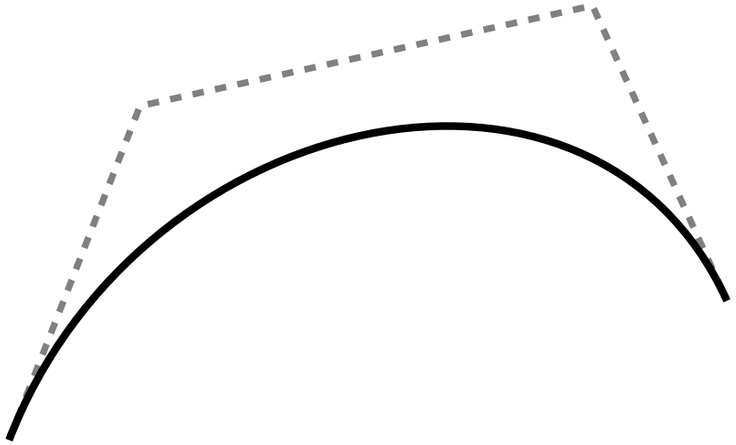
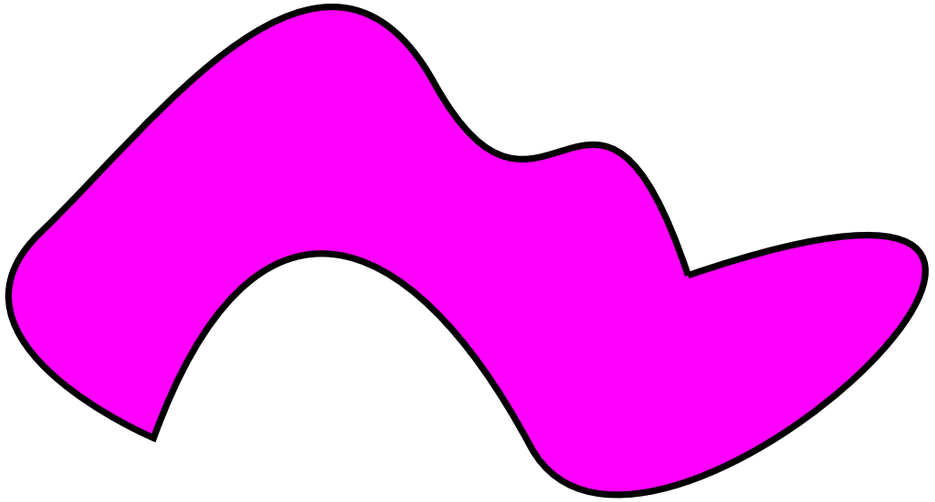
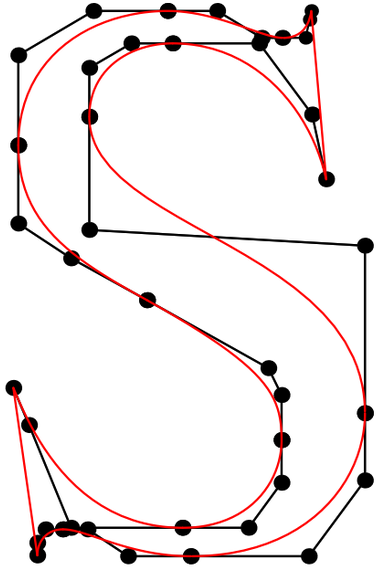
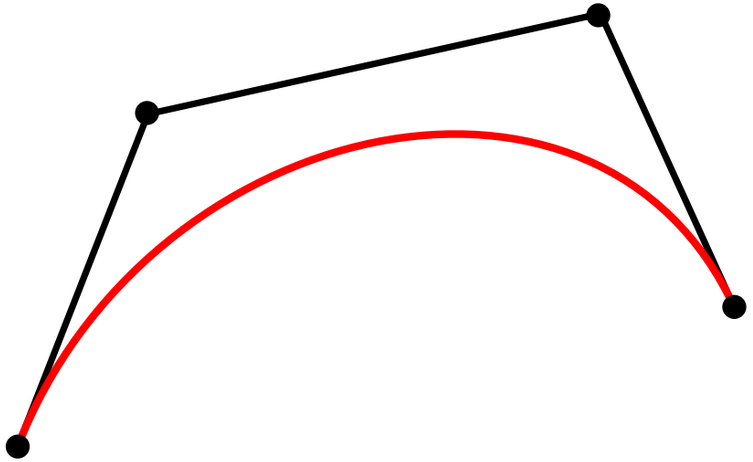


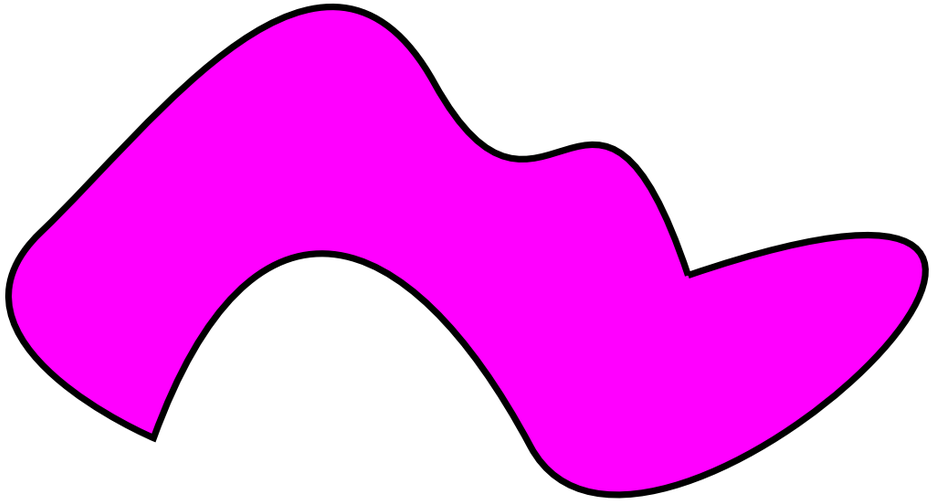


Vector Graphics

- Vector graphics primitives, with attributes
  - Lines, polylines, polygons
  - Circles, ellipses
  - **Bézier curves (cubics), bezigons, fonts (Inkscape demo)**
  - Diffusion and shading curves
  - Color gradients







Vector Graphics

- Vector graphics primitives, with attributes
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  - Circles, ellipses
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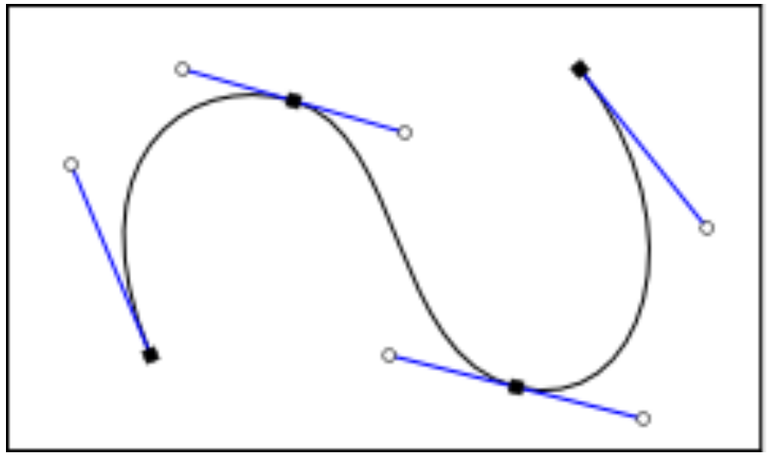
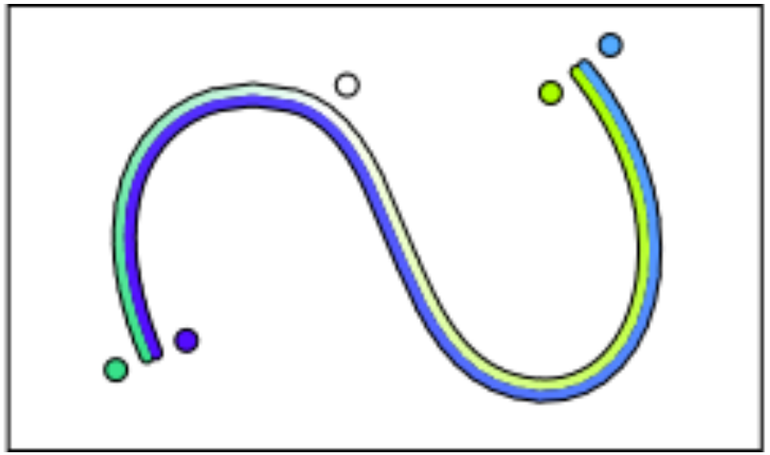
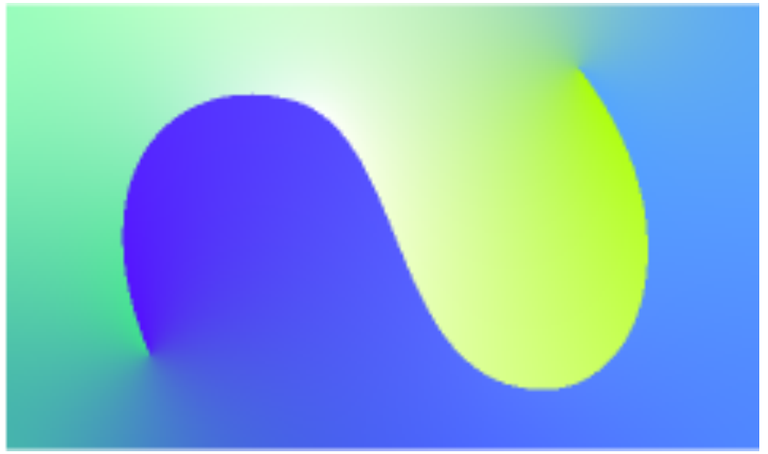
  - **Diffusion** and shading **curves**
  - Color gradients



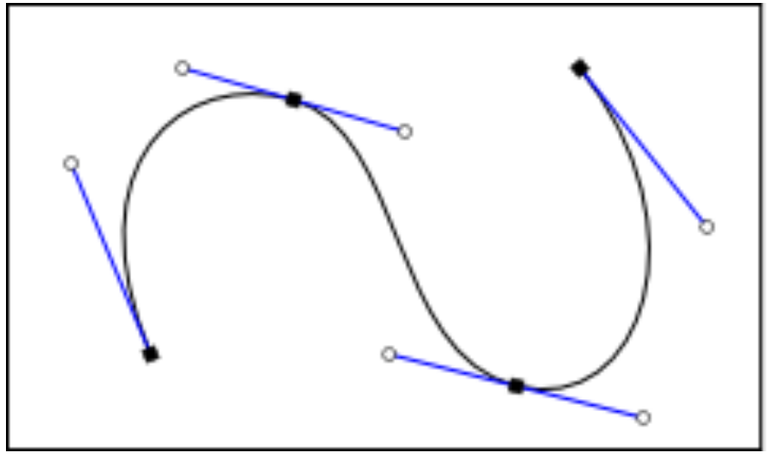


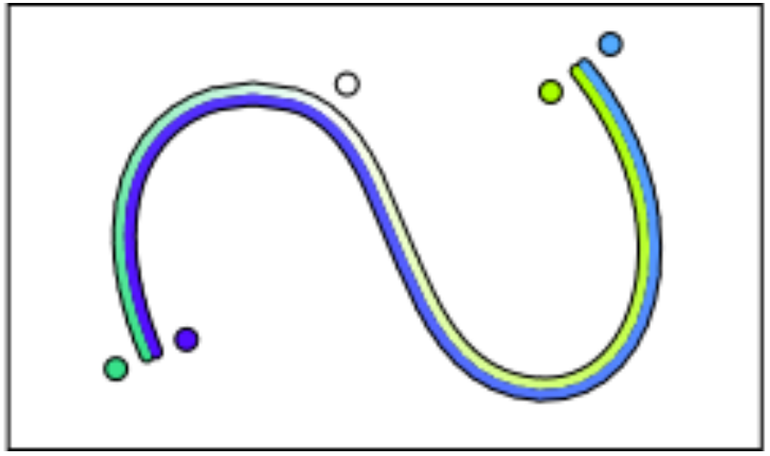


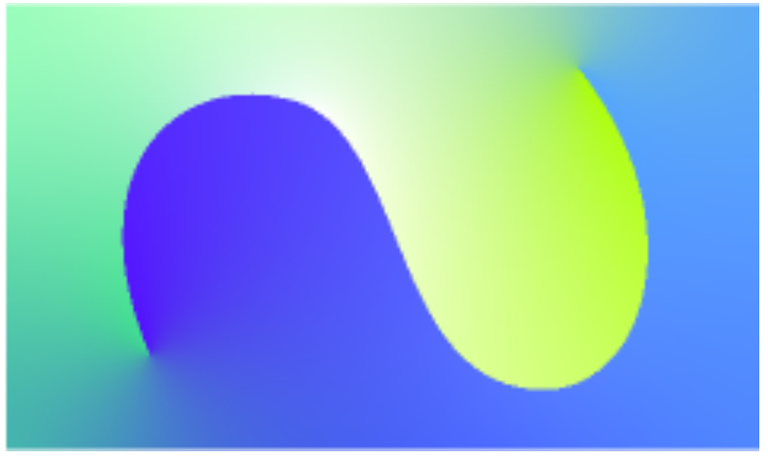
[Orzan et al. 2008]

Vector Graphics

- Vector graphics primitives, with attributes
  - Lines, polylines, polygons
  - Circles, ellipses
  - Bézier curves (cubics), bezigons, fonts
  - **Diffusion** and shading **curves (browser demo)**
  - Color gradients







[Orzan et al. 2008]

Vector Graphics

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  - Bézier curves (cubics), bezigons, fonts
  - **Diffusion** and shading **curves**
  - Color gradients

[Finch et al. 2011]

Vector Graphics

- Vector graphics primitives, with attributes
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  - Circles, ellipses
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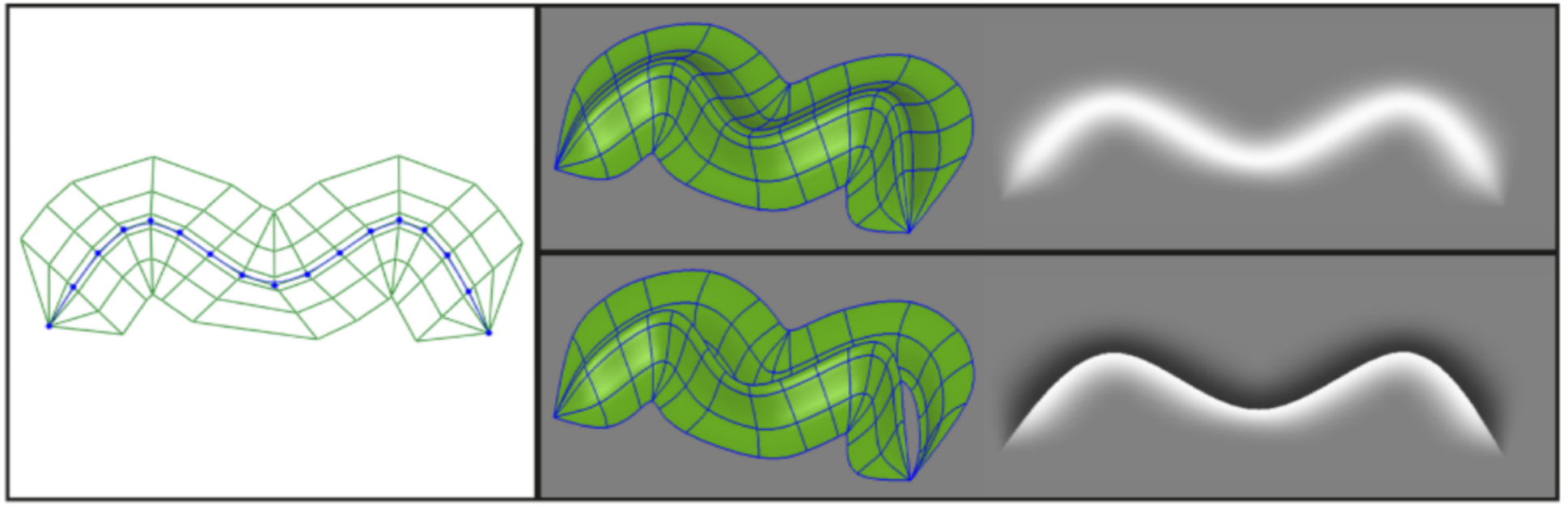

[Lieng et al. 2015]

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  - Circles, ellipses
  - Bézier curves (cubics), bezigons, fonts
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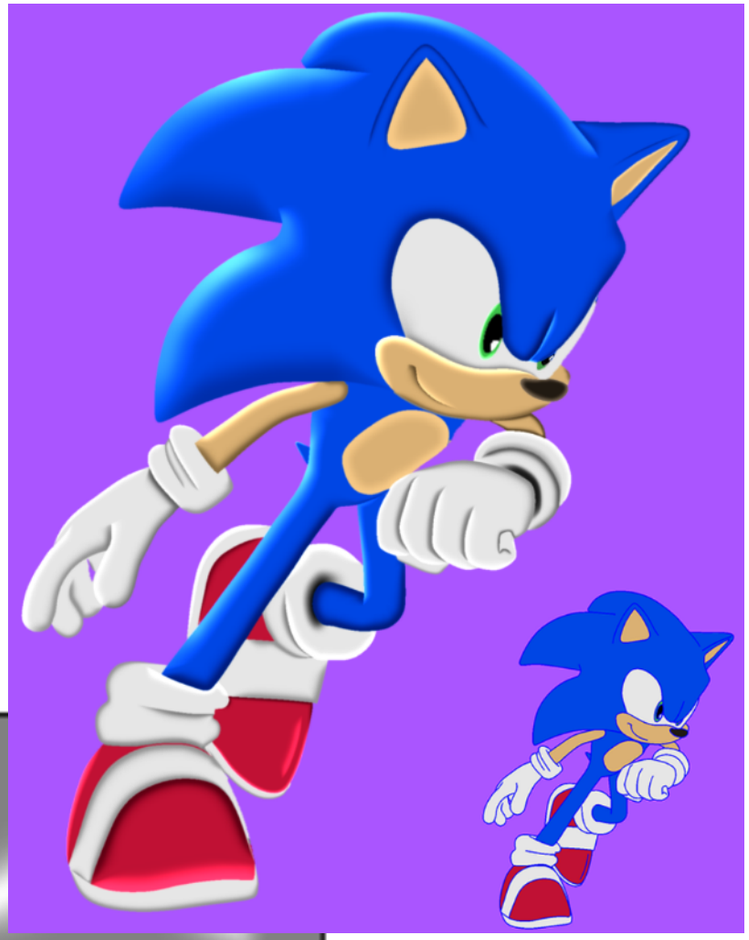

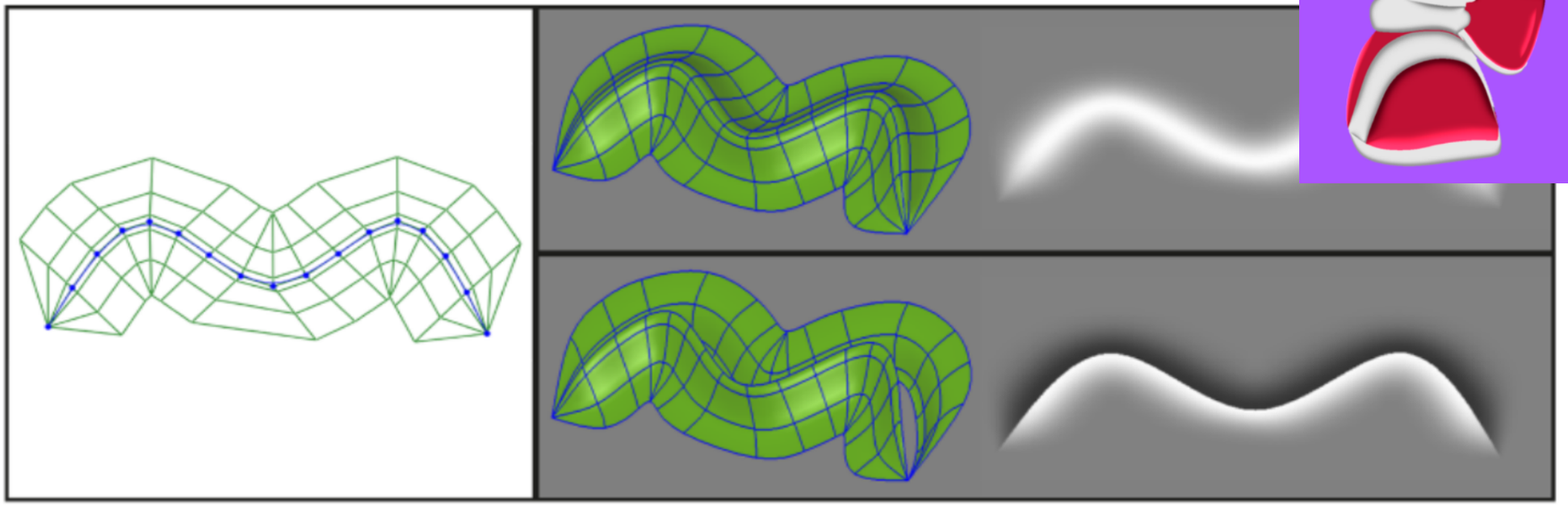

[Lieng et al. 2015]

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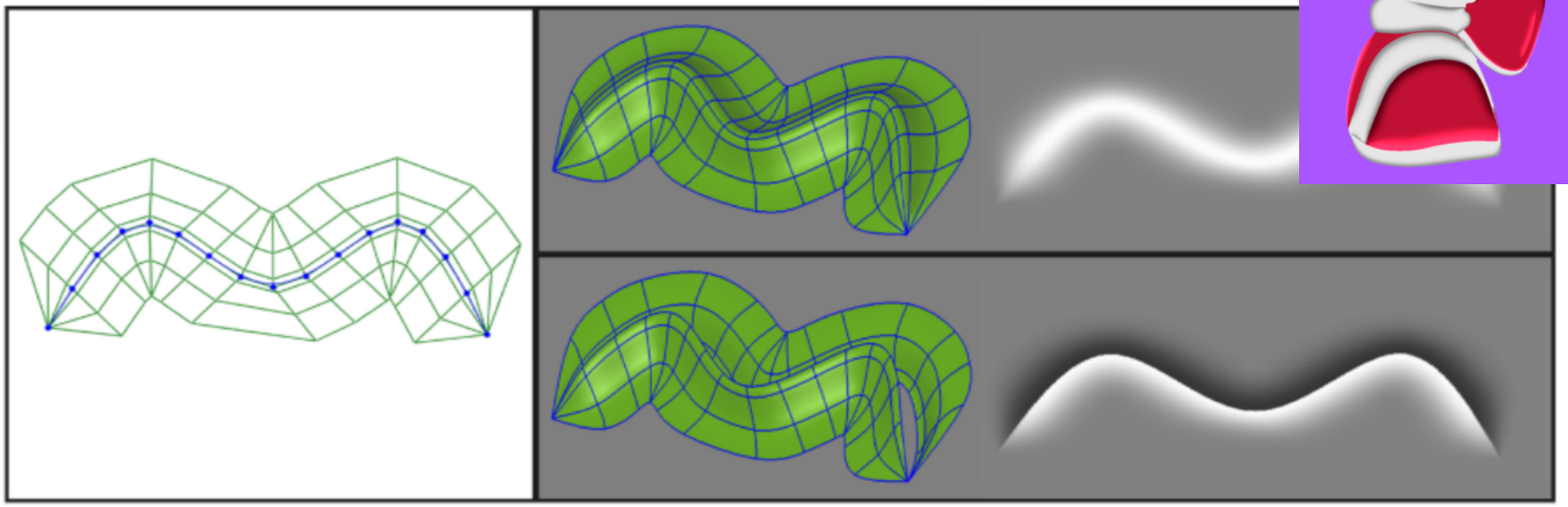

[Lieng et al. 2015]

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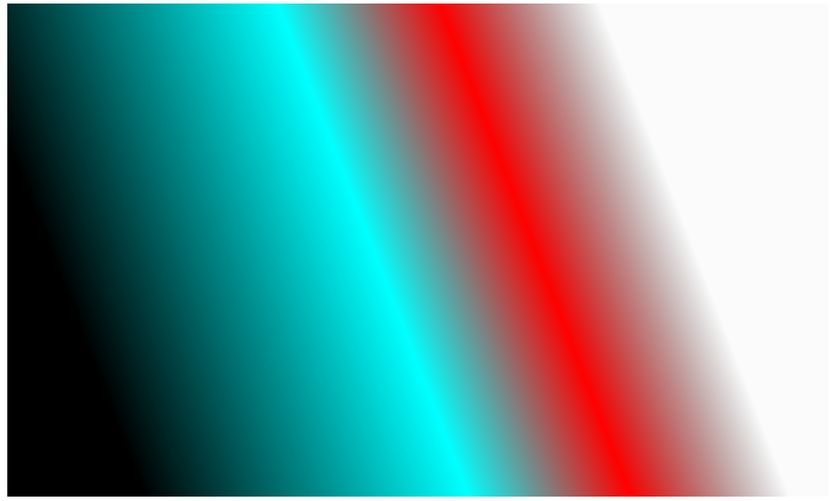

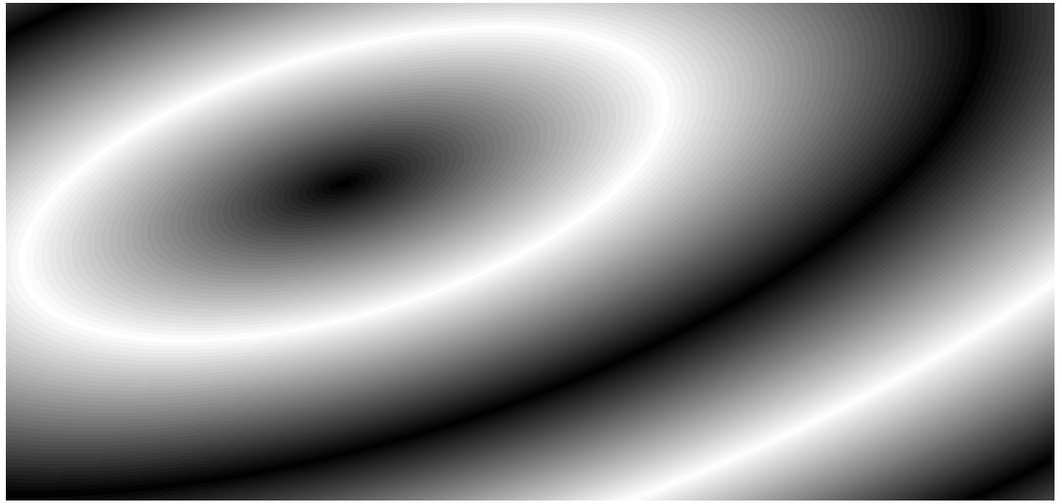

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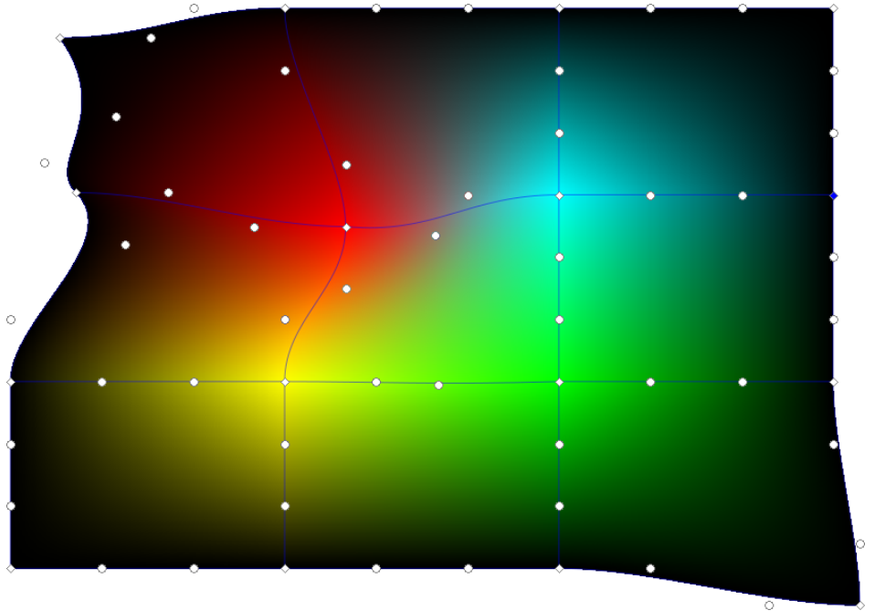

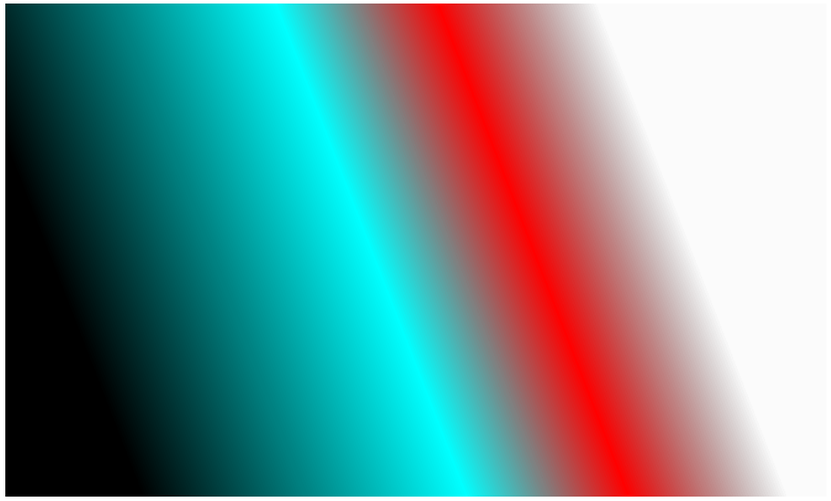

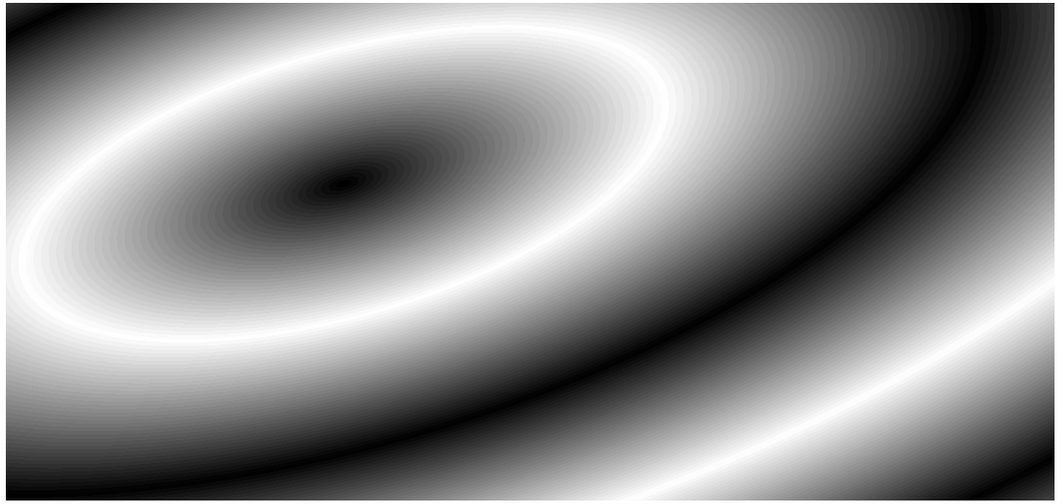

Vector Graphics

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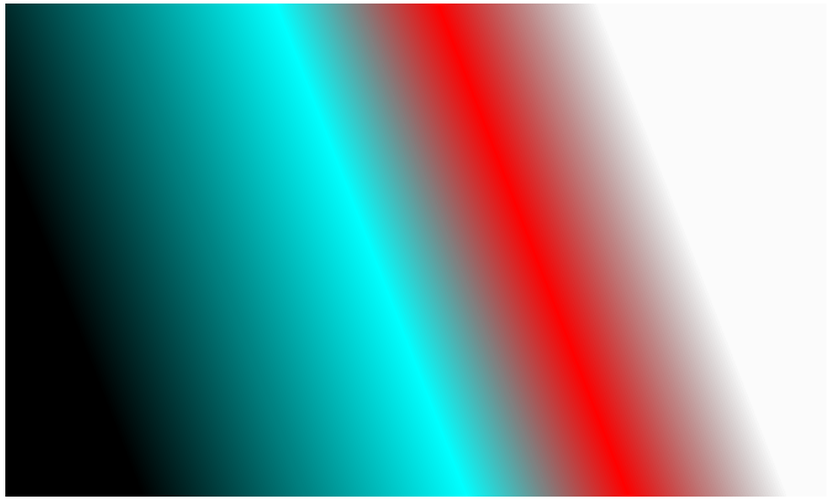
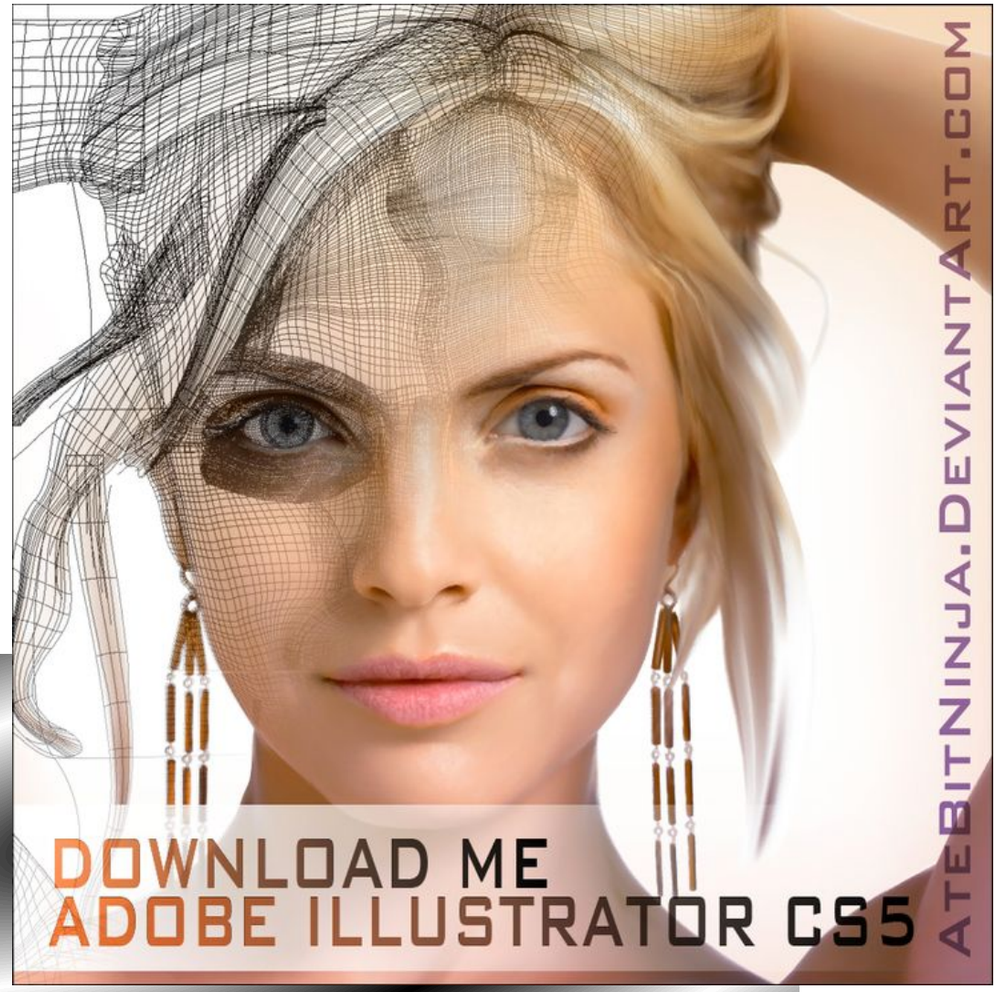
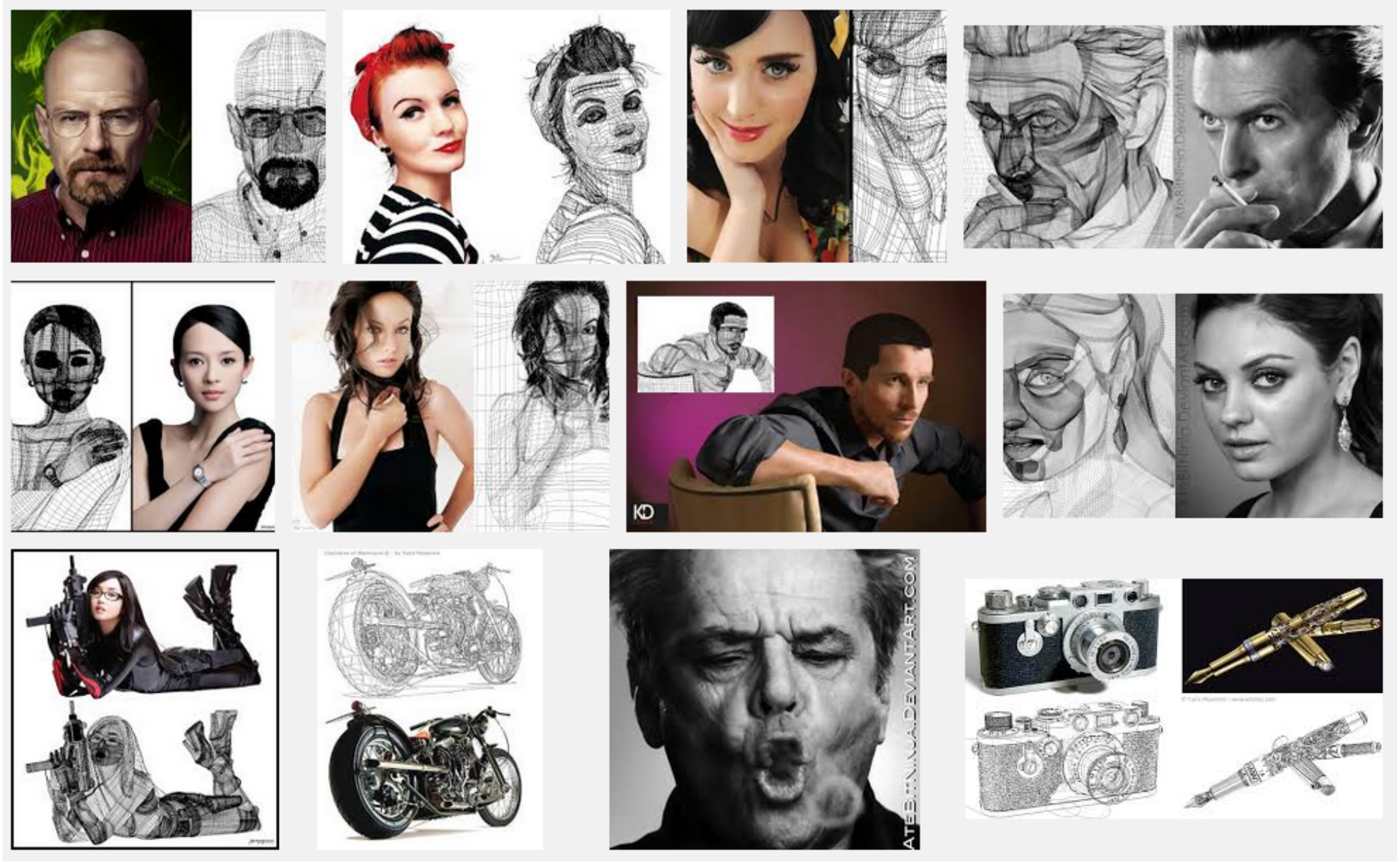


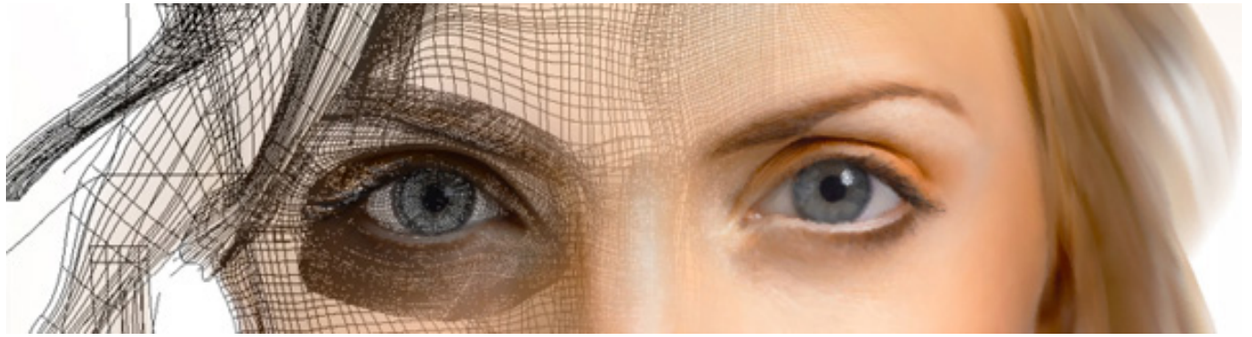


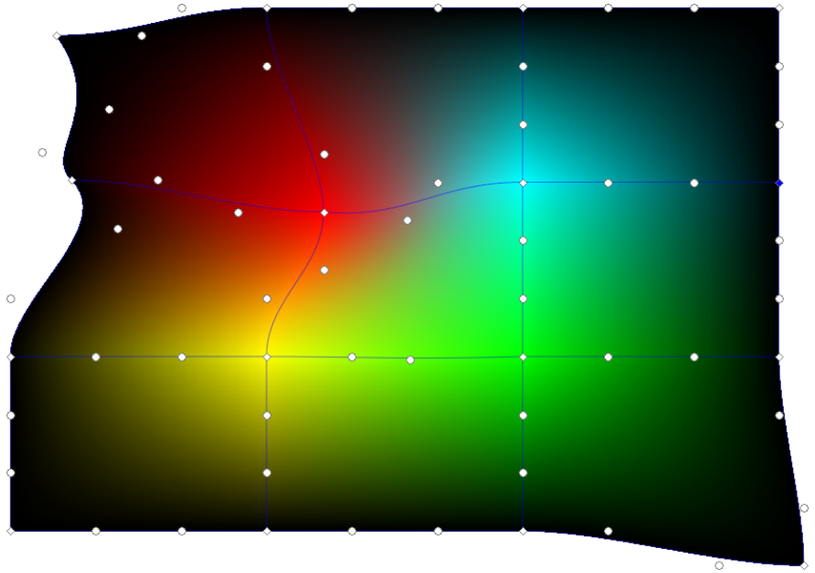
Gradient Mesh

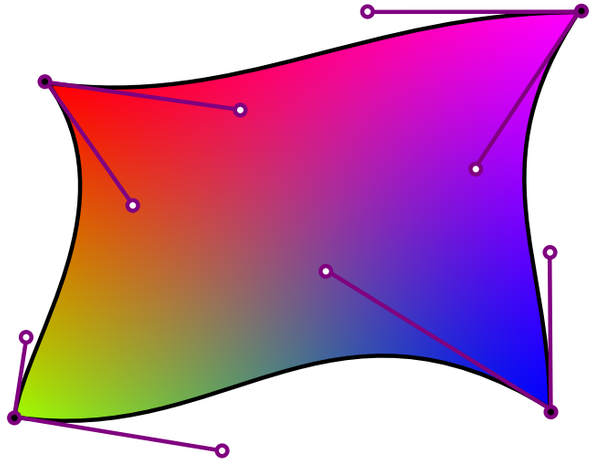


[Google images]

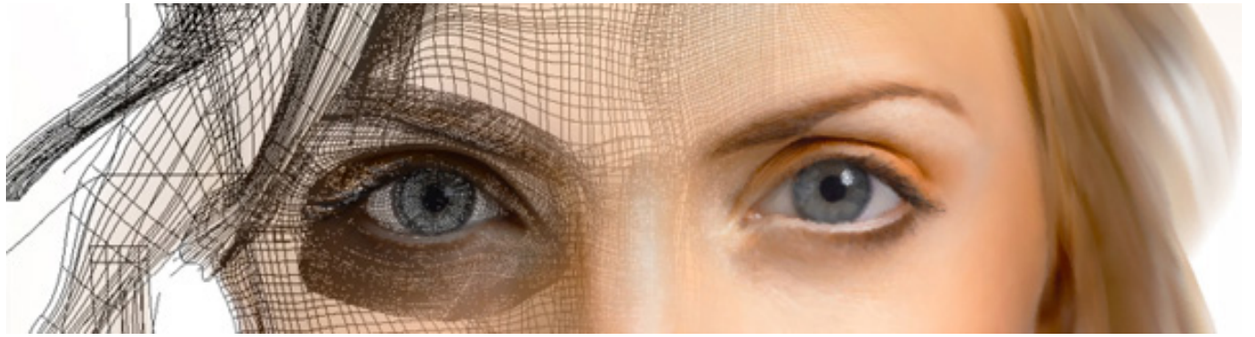
Gradient Mesh

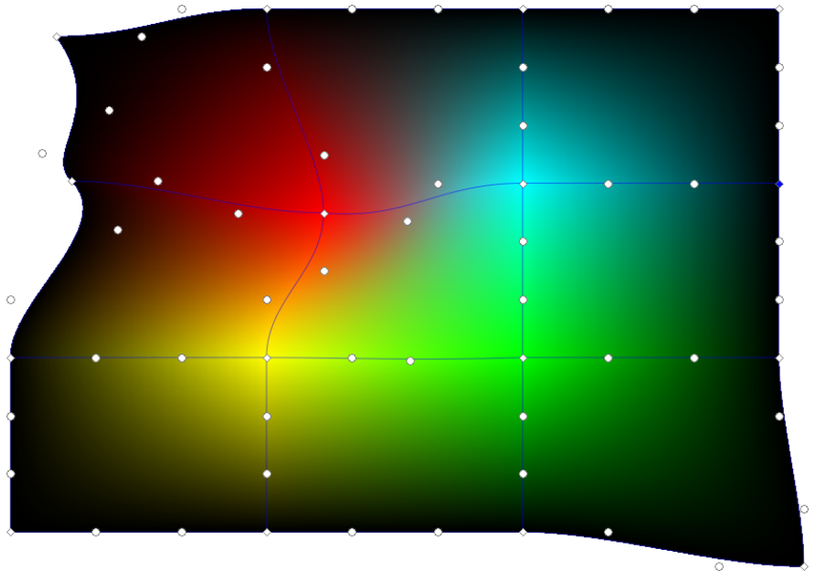


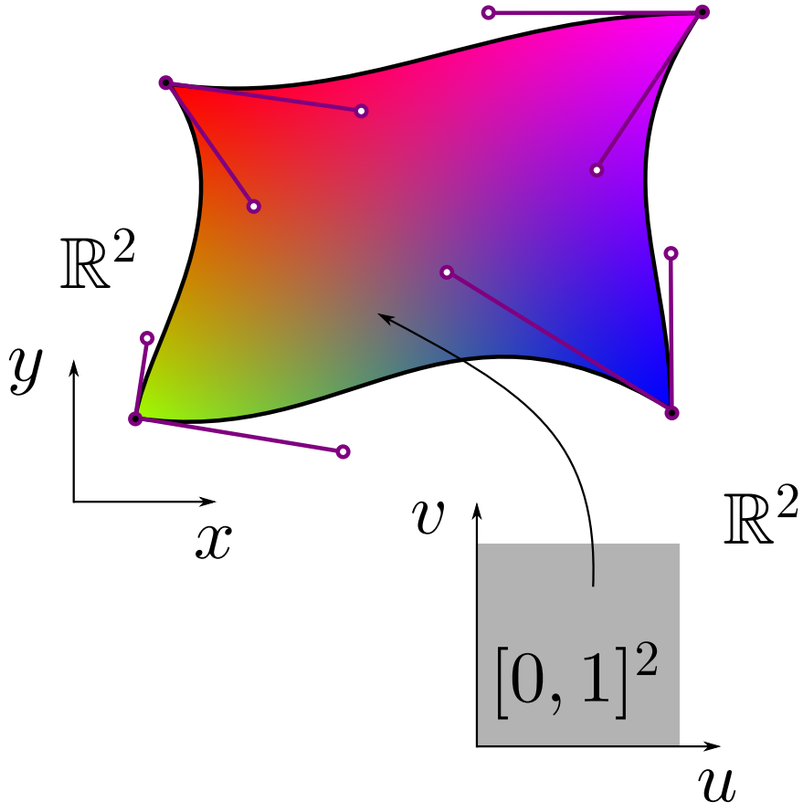




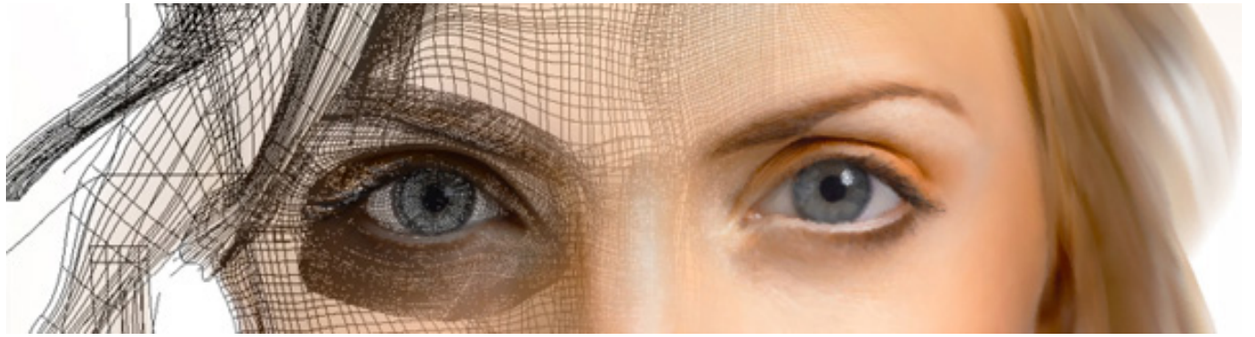
Gradient Mesh

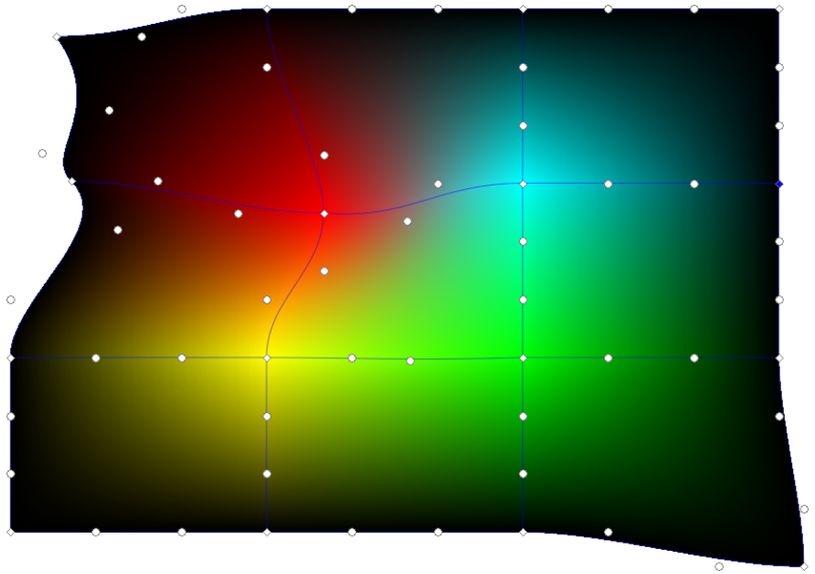


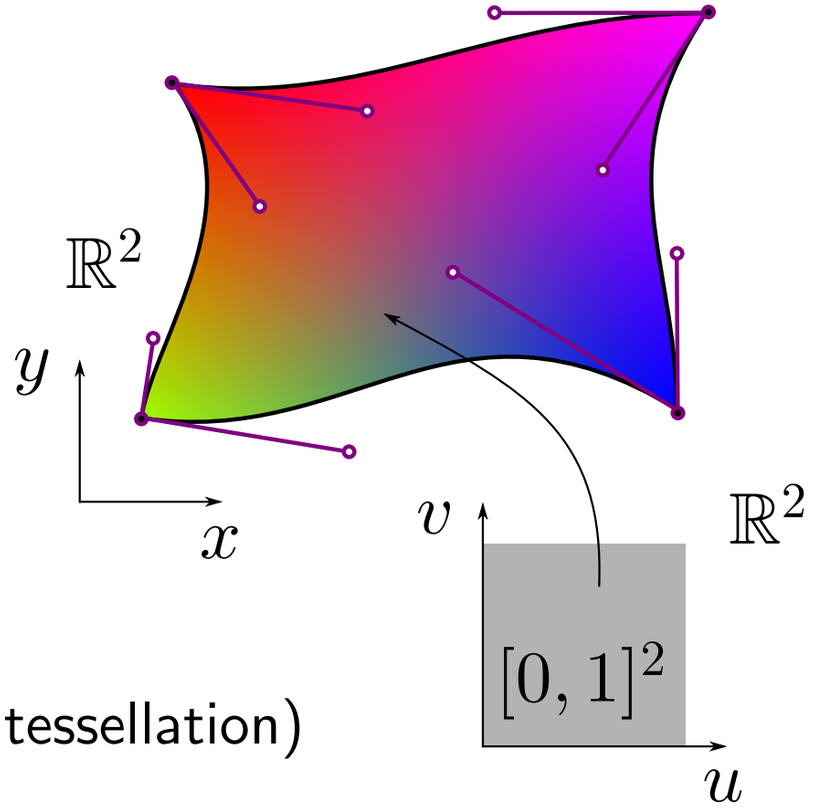




Gradient Mesh







Demo (with tessellation)

Gradient Mesh

- Gradient mesh of Adobe Illustrator/Corel Draw/Inkscape
  - Rectangular array of Ferguson patches
  - C^1 continuity
  - Used by artists to create stunning vector graphics
  - Often several meshes are used (in layers)
  -
  -

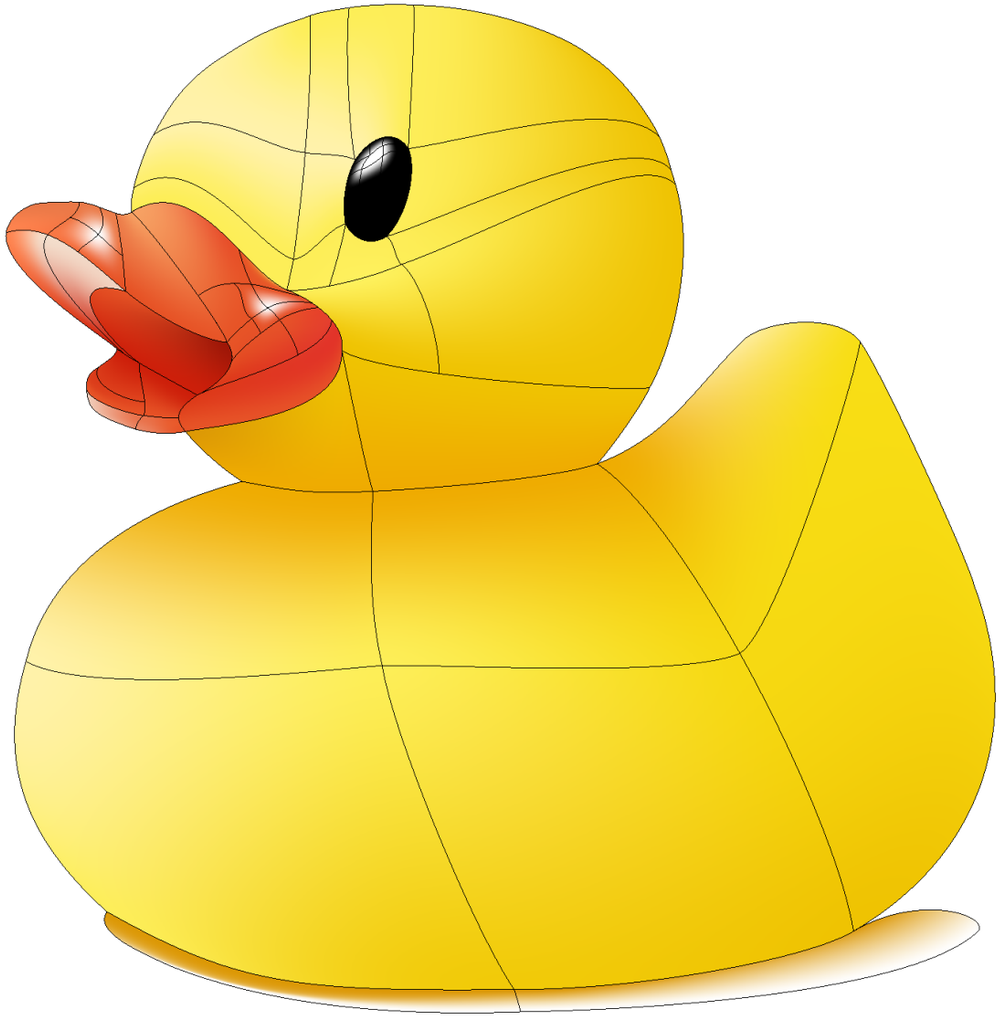
[Ruslan Khasanov]

Gradient Mesh

- Gradient mesh of Adobe Illustrator/Corel Draw/Inkscape
  - Rectangular array of Ferguson patches
  - C^1 continuity
  - Used by artists to create stunning vector graphics
  - Often several meshes are used (in layers)
  - **Topology restricted**
  - and thus too rigid for (automatic) vectorization

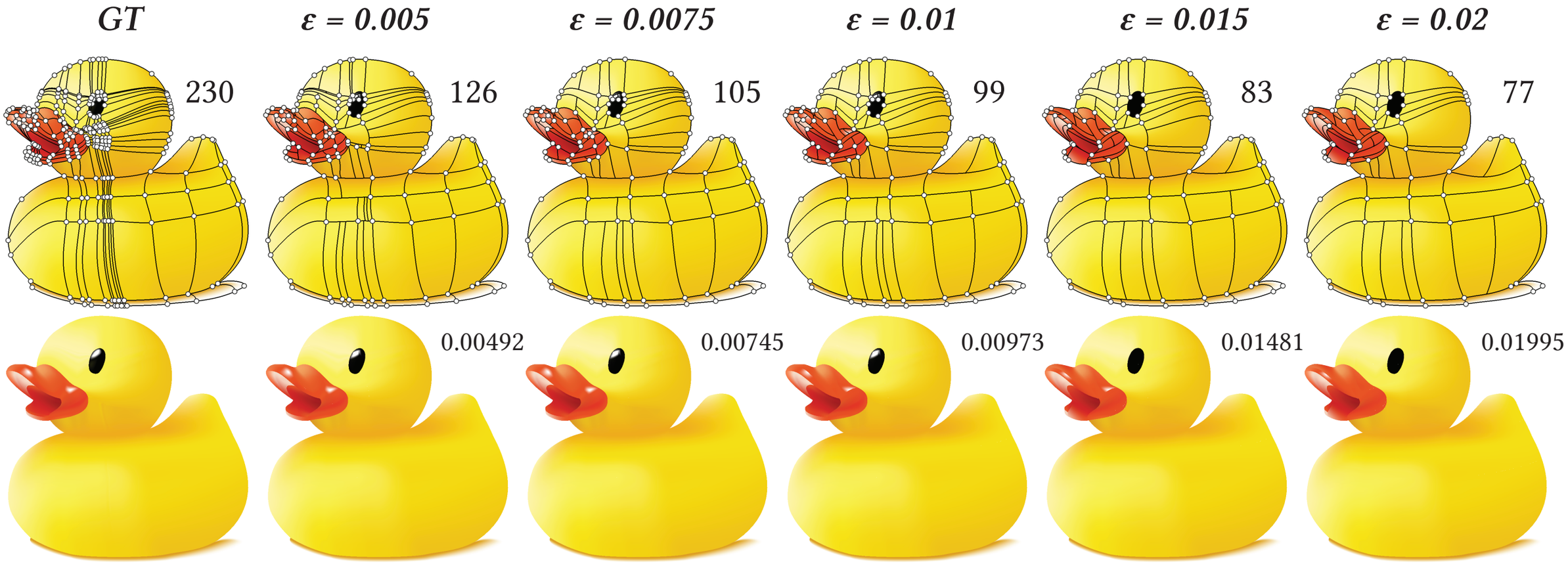
[Ruslan Khasanov]

Gradient Mesh: Refinement



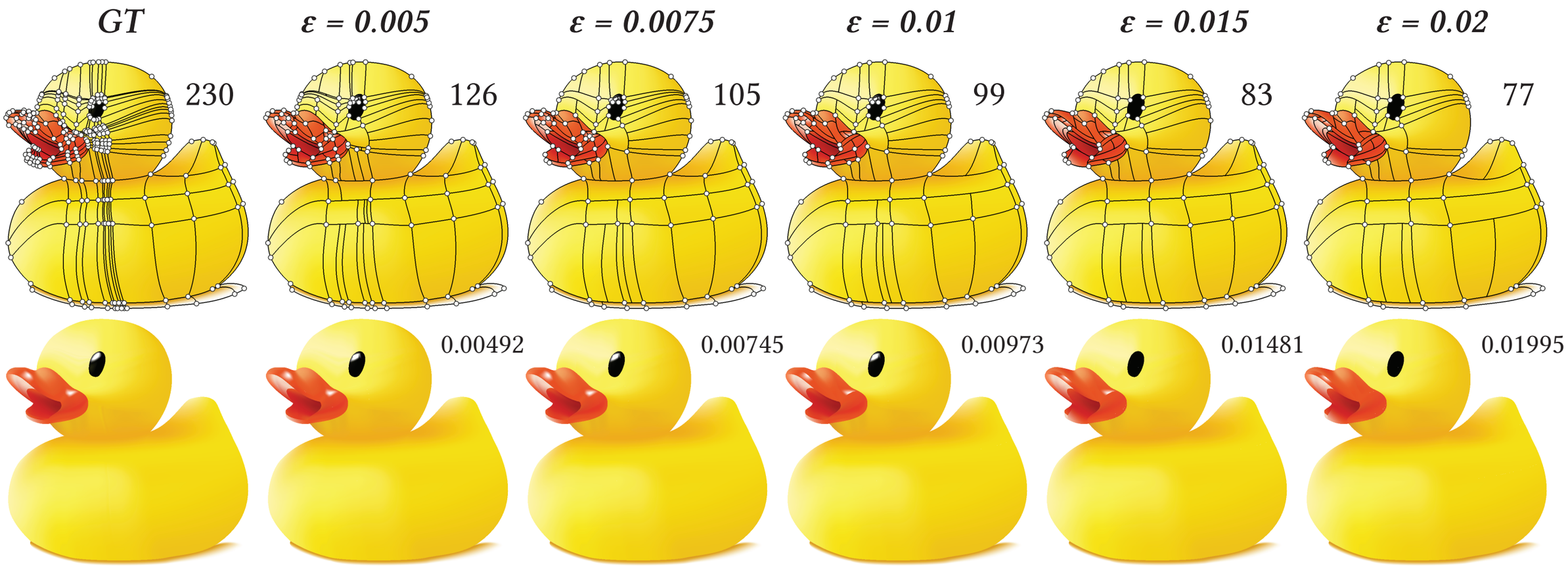
Local refinement (and branching) [Barendrecht et al. (2018)] - Demo

Gradient Mesh: Simplification/Coarsening

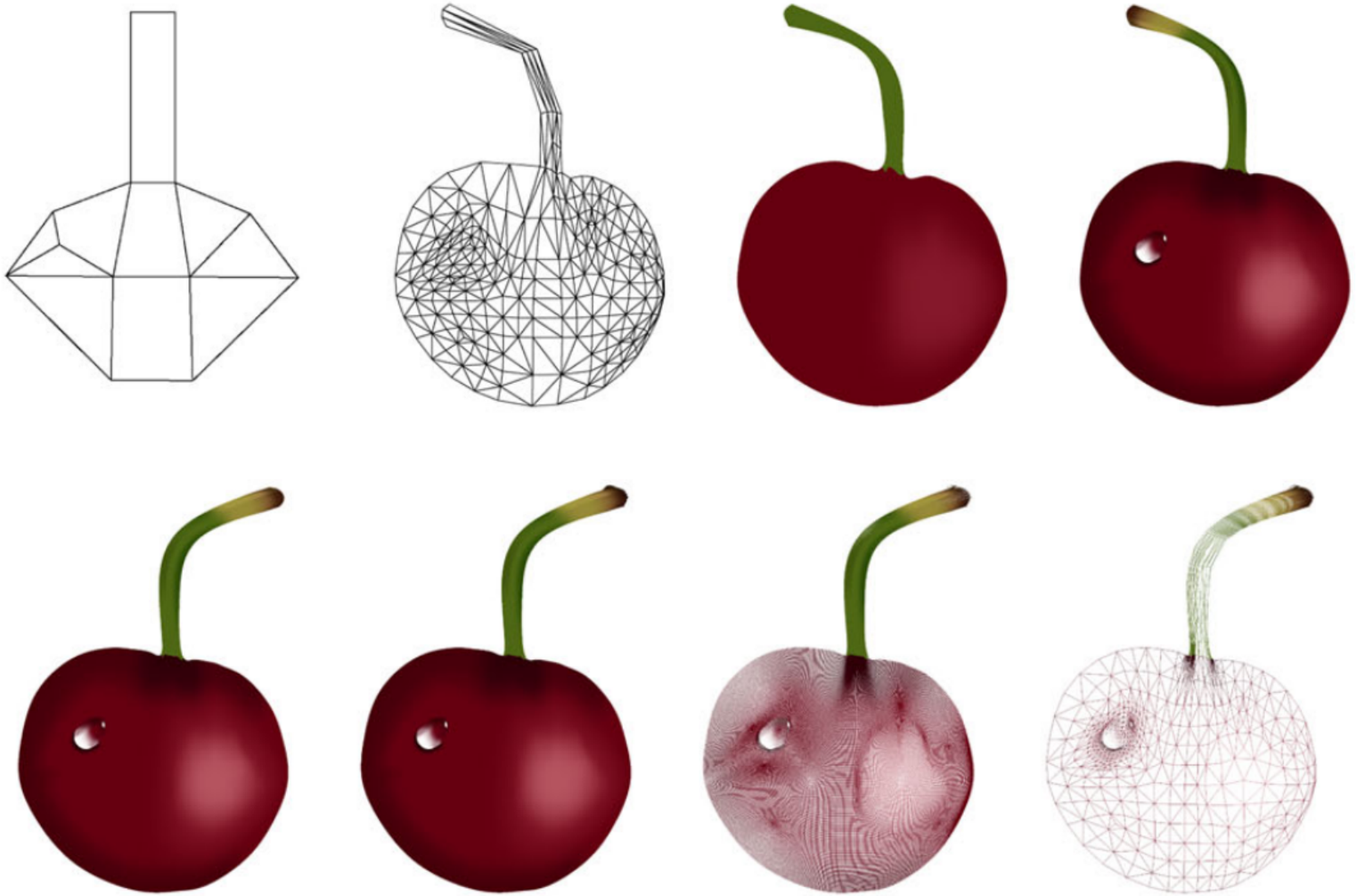


ꟻLIP and other error metrics [Kato et al. 2026]

Gradient Mesh: Simplification/Coarsening

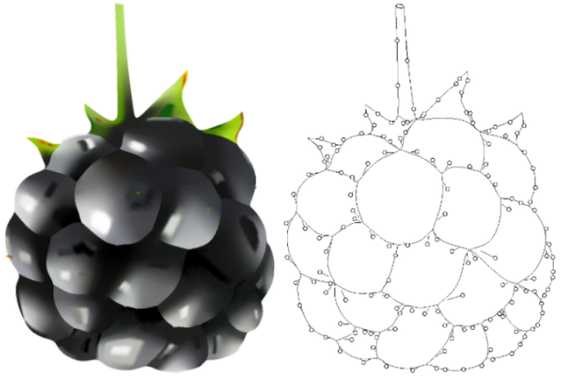


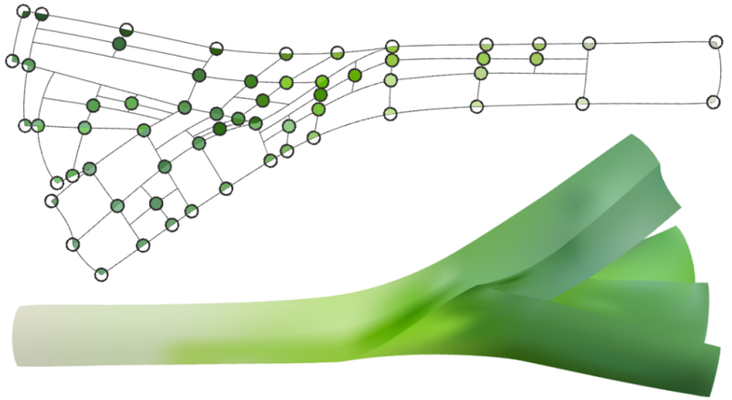
Multiresolution Editing

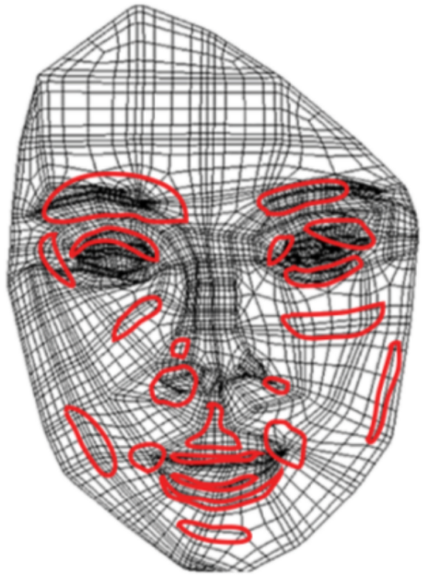


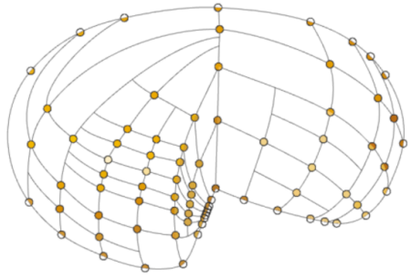
Global subdivision (78k polygons) vs. our method (2k polygons) [Zhou et al. 2022]; leek demo

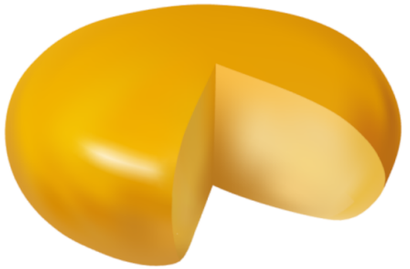
Results





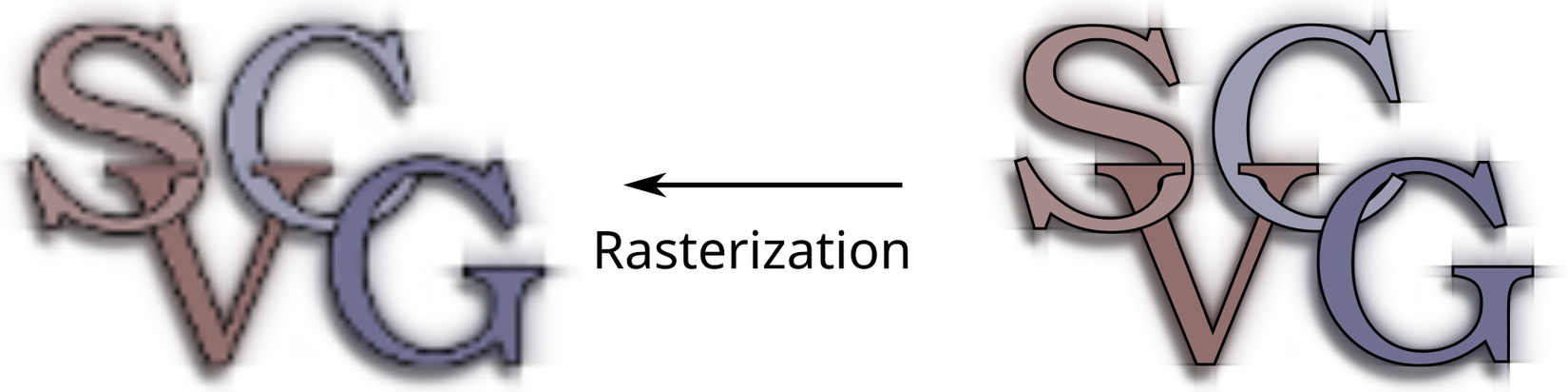






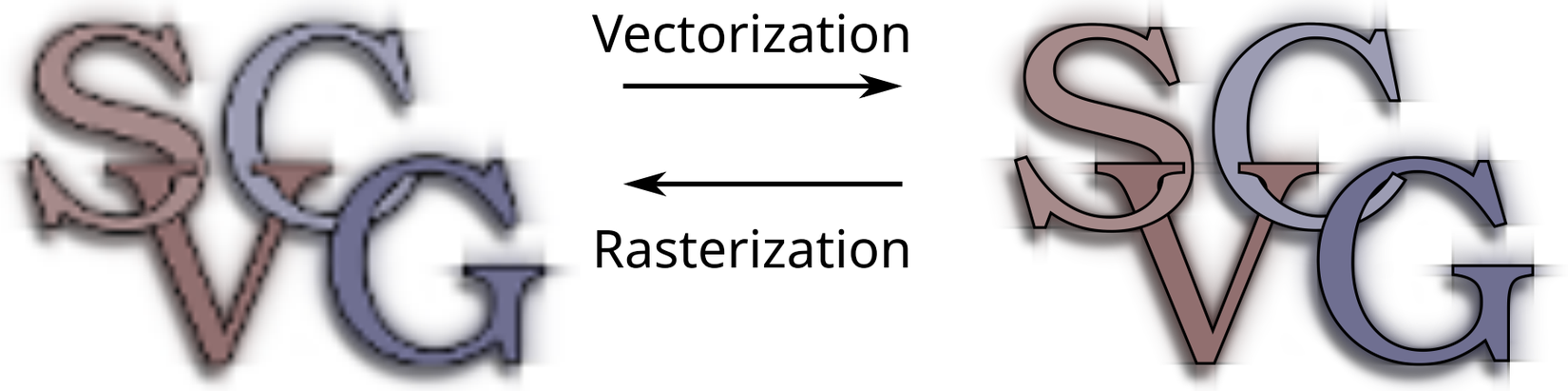
Rasterization vs. Vectorization

- Rasterization
  - For display, vector images need to be rasterized
  - This is typically simple and (very) fast
-
  -
  -



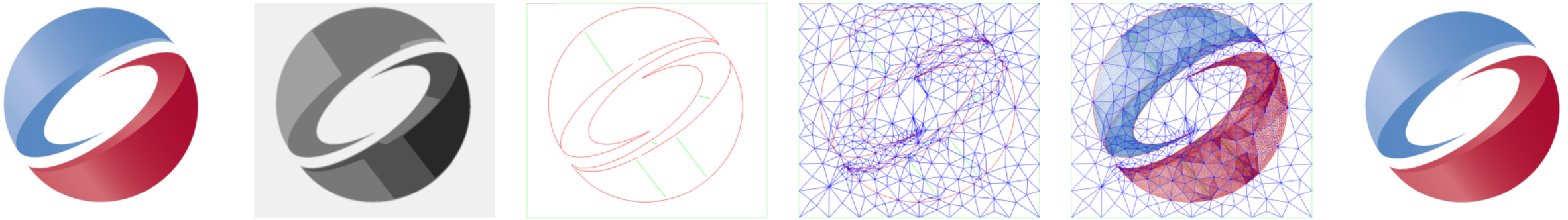
Rasterization vs. Vectorization

- Rasterization
  - For display, vector images need to be rasterized
  - This is typically simple and (very) fast
- Vectorization
  - Convert a raster image to a vector image
  - Difficult; still an open problem



Automatic Vectorization

- Pipeline:
  - Detect salient edges in the image (e.g. using Canny detector), represented as pixel chains
  - Fit a Bézier curve (or spline) to each chain (least squares fitting)
  - Mesh the entire image using triangles or quads (with curved edges; Bézier guarding)
  - respecting the found and vectorized edges
  - Assign colors to vertices by fitting

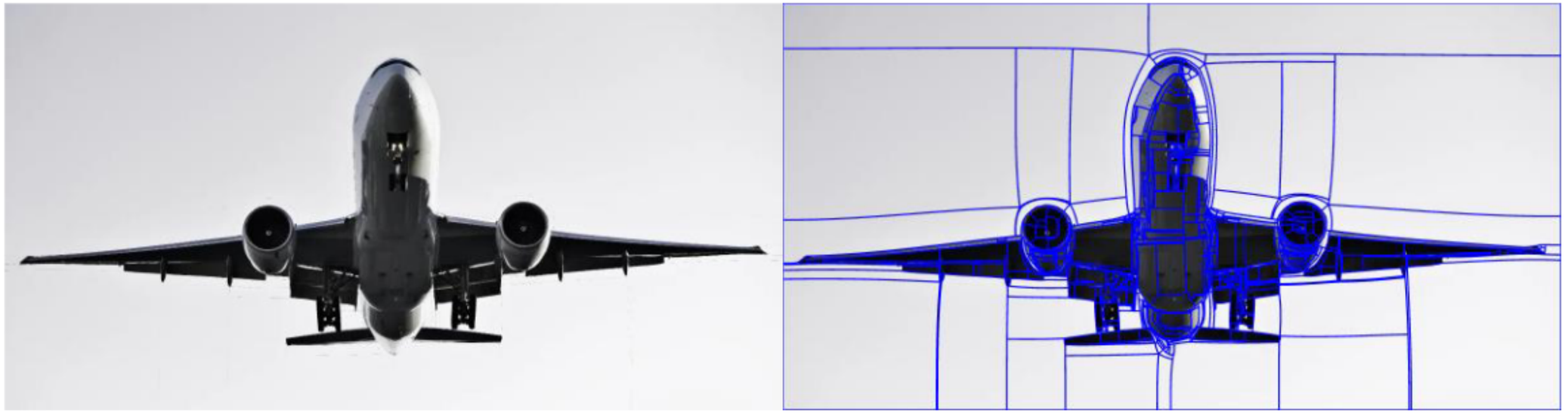


[Hettinga et al. 2022]

Automatic Vectorization

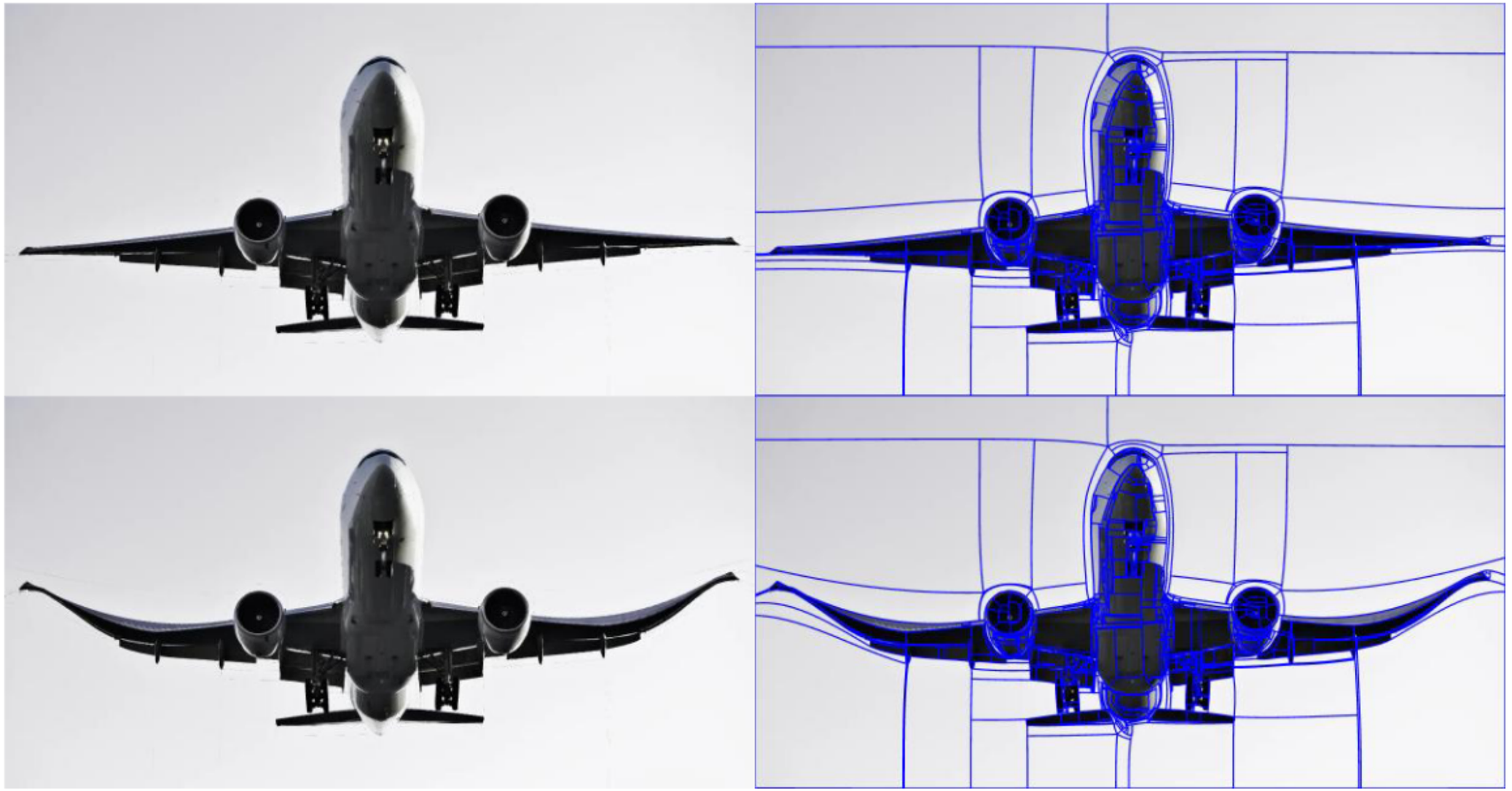
Triangular mesh [Hettinga et al. 2022]

Automatic Vectorization



Quad mesh (Coons patches) [He et al. 2024]

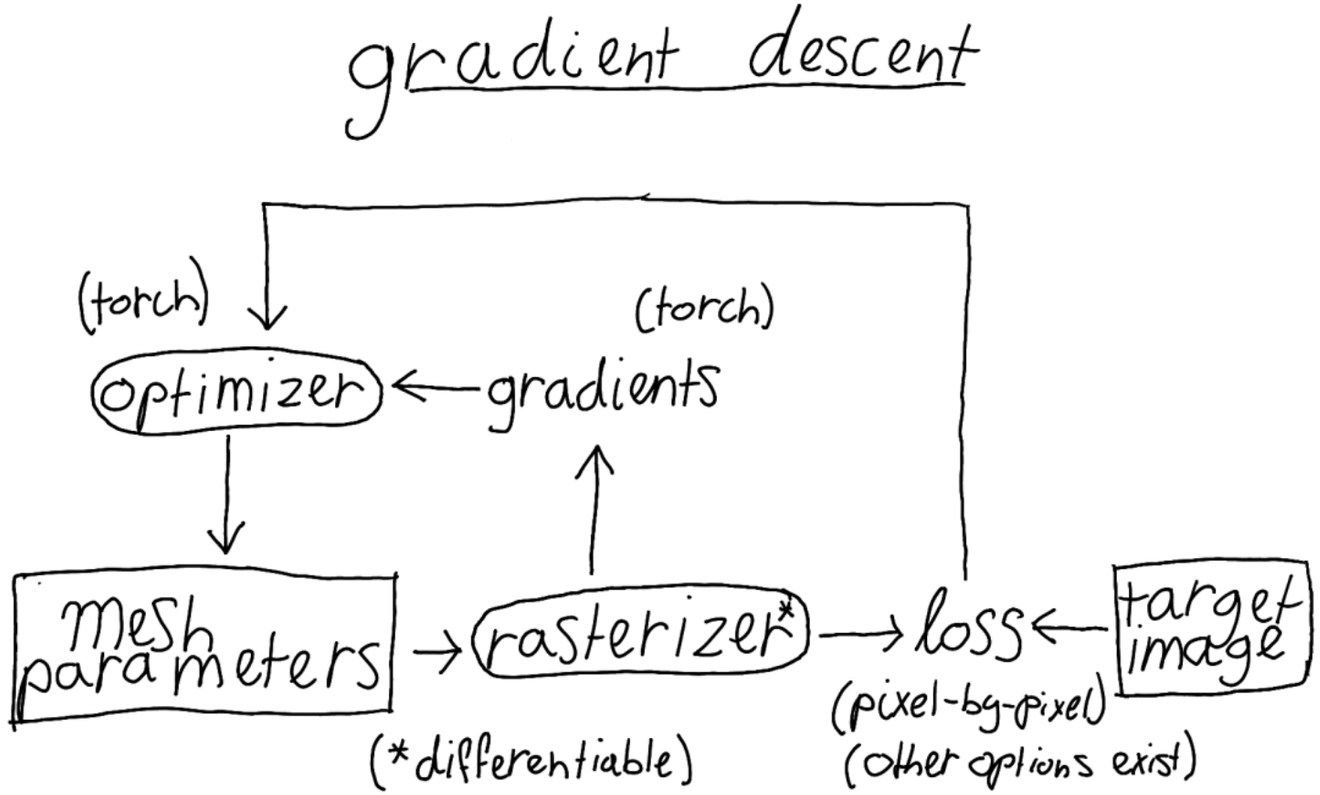
Automatic Vectorization



Quad mesh (Coons patches) [He et al. 2024]

(Gradient) Mesh Optimization

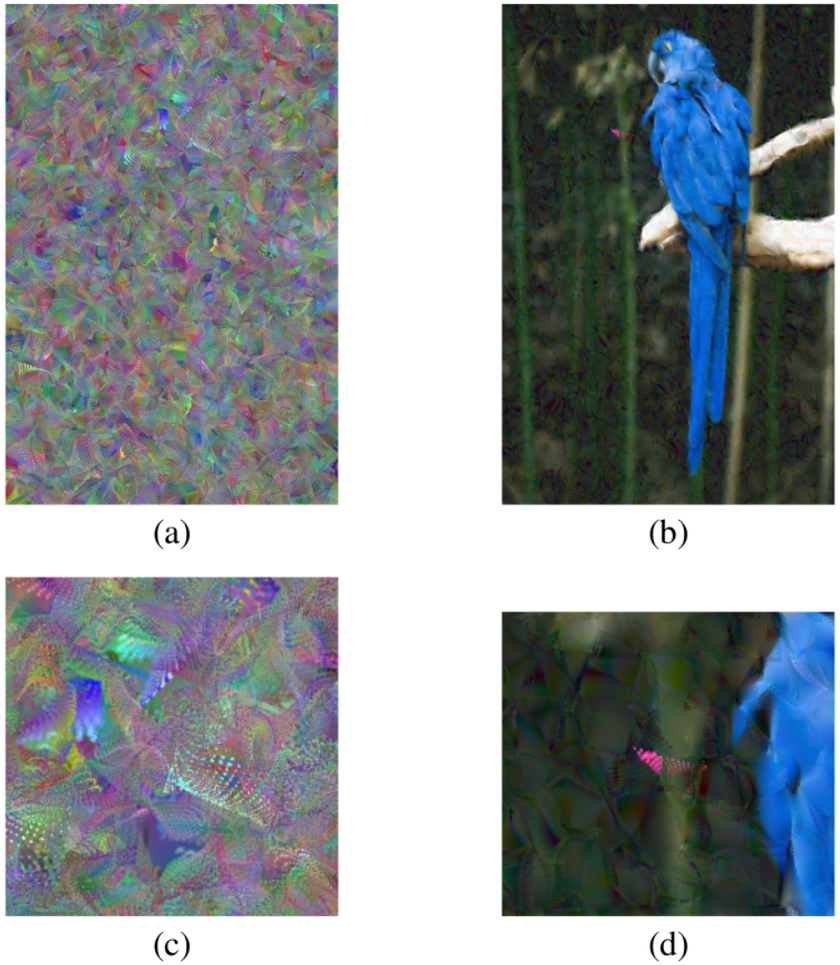
- Differentiable rasterizer: Tzu-Mao et al. 2020
-



[He et al. 2026]

(Gradient) Mesh Optimization

- Differentiable rasterizer: Tzu-Mao et al. 2020
- Naive initialization

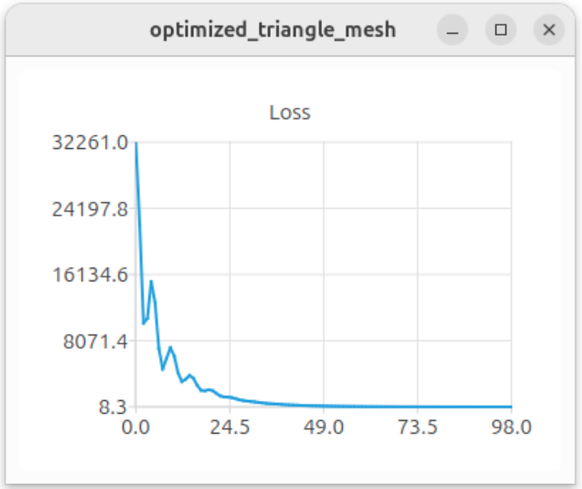


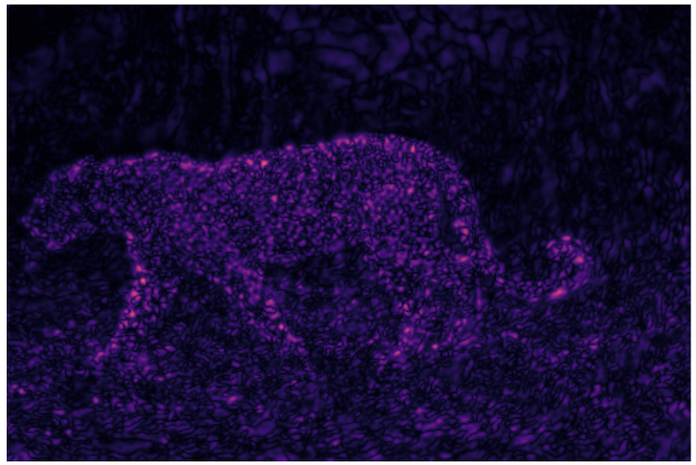
(a) (b) (c) (d)

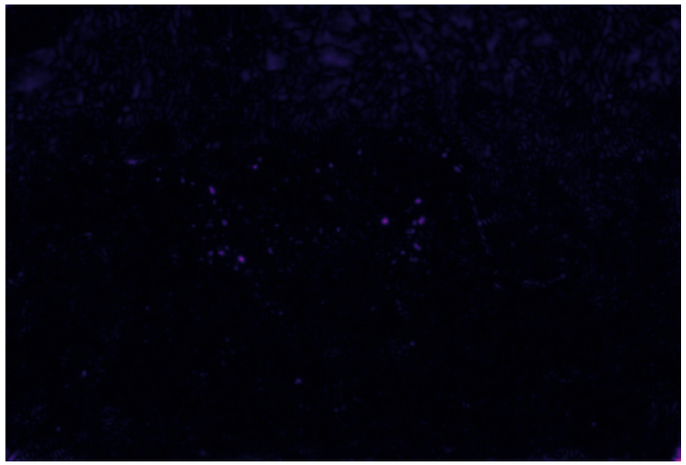
[Veenhuijsen, 2023; **videos**]

(Gradient) Mesh Optimization

- ~~Differentiable rasterizer~~ Monte Carlo integration
- Initialization via our previous method





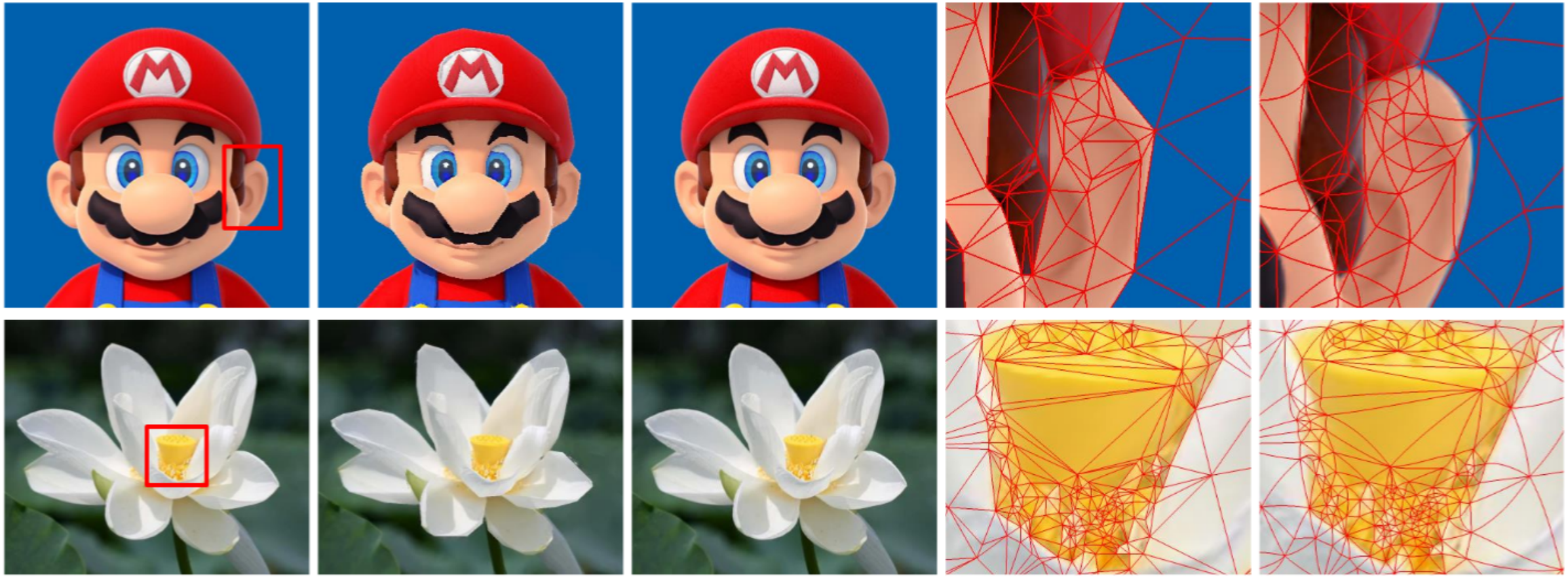


[He at al. 2026; **gif**]

(Gradient) Mesh Optimization

- Minimize $\epsilon(\theta) = \sum_{i=1}^{n_f} \int_{\Omega_f} \| I_S(x,y) - R_f(x,y;\theta_f) \|^2 \, \mathrm{d}x\mathrm{d}y$
  - I_S: continuous color surface (bilinear interpolation)
  - $R_f(x,y;\theta_f)$: color value at (x,y) within the range Ω_f of triangle f based on its parameters θ_f
- Change of coordinates: $\epsilon(\theta) = \sum_{i=1}^{n_f} \int_{\triangle} \| I_S(\mathbf{x}) - R_f(\mathbf{x}) \|^2 \, J_f(\mathbf{u}) \, \mathrm{d}\mathbf{u}$
  - $\mathbf{x} = (x(\mathbf{u}), y(\mathbf{u}))$
  - $J_f(\mathbf{u})$: Jacobian determinant
- Monte Carlo: $\epsilon(\theta) \approx \epsilon_I(\theta) = \sum_{i=1}^{n_f} \frac{1}{2s_f} \sum_{j=1}^{s_f} \| I_S(P_f(\mathbf{u}_j)) - C_f(\mathbf{u}_j) \|^2 \, J_f(\mathbf{u}_j)$
  - $P_f(\mathbf{u}_j)$: position
  - $C_f(\mathbf{u}_j)$: color

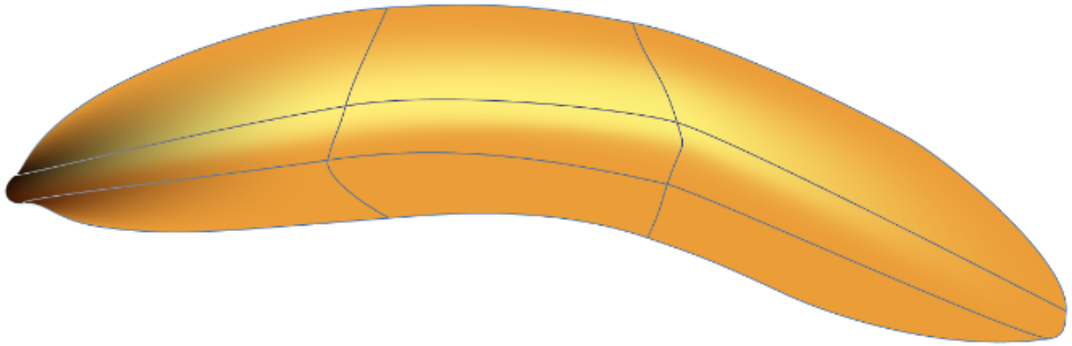
(Gradient) Mesh Optimization

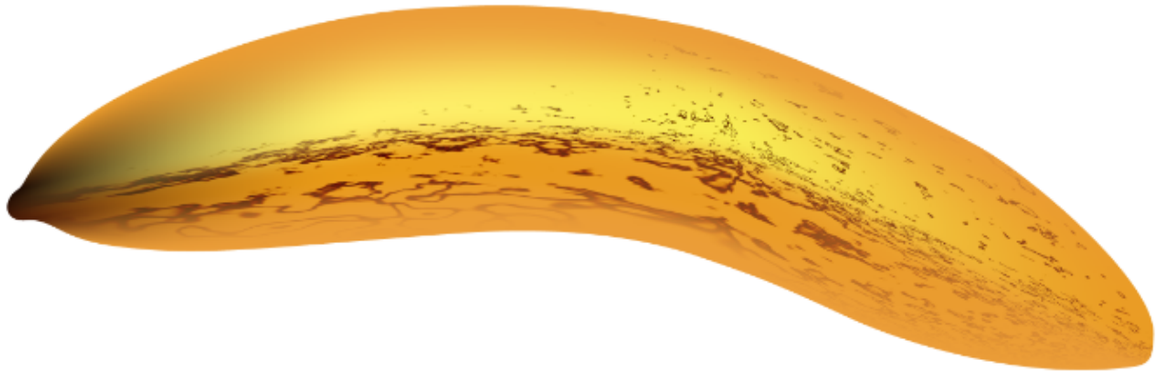


[Fang et al. 2023; He at al. 2026]

Where Next?

- Unification (diffusion curves, gradient meshes, ...)
- Standardization (SVG, PDF, browsers, tools, ...)
- (Procedural) **textures and noise** - modeling demo
- Video vectorization
- Role of AI?

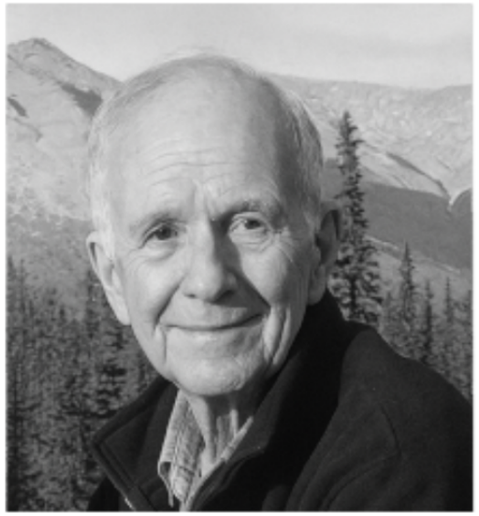




[Hettinga et al. 2019]

Where Next?

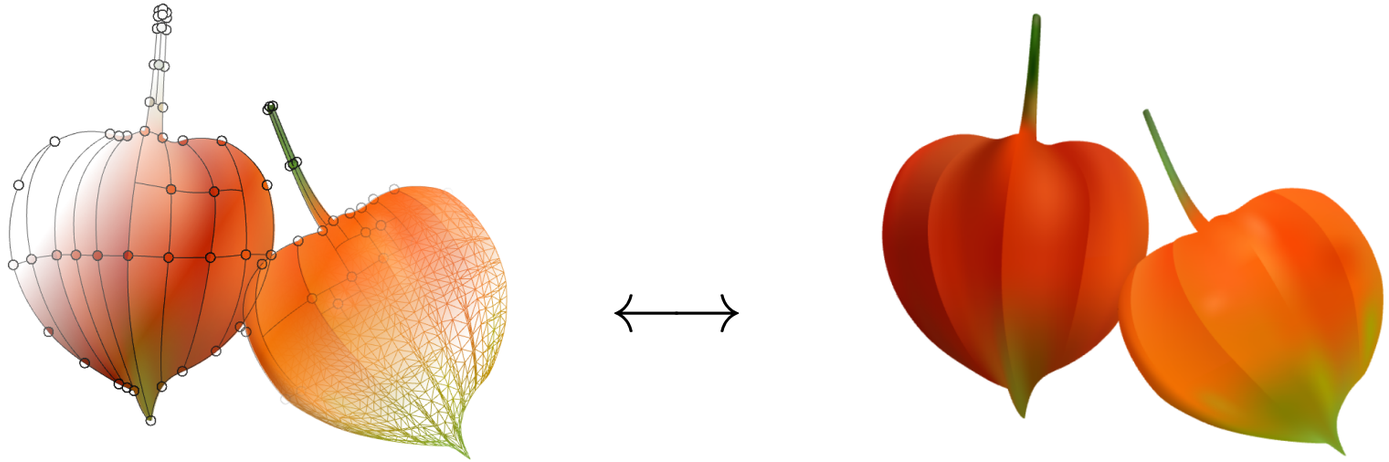
- Unification (diffusion curves, gradient meshes, ...)
- Standardization (SVG, PDF, browsers, tools, ...)
- (Procedural) textures and noise
- Video vectorization
- **Role of AI?**



[K. 2056]

Summary

- Vector graphics
  - Resolution independent, fully scalable
  - Editable, smaller file size
- Primitives
  - Lines, triangles, radial/linear gradients
  - Circles, quadratic/cubic curves
  - Diffusion curves and **gradient meshes**
- Gradient meshes and beyond
  - Local/hierarchical refinement
  - Image vectorization
  - Optimization



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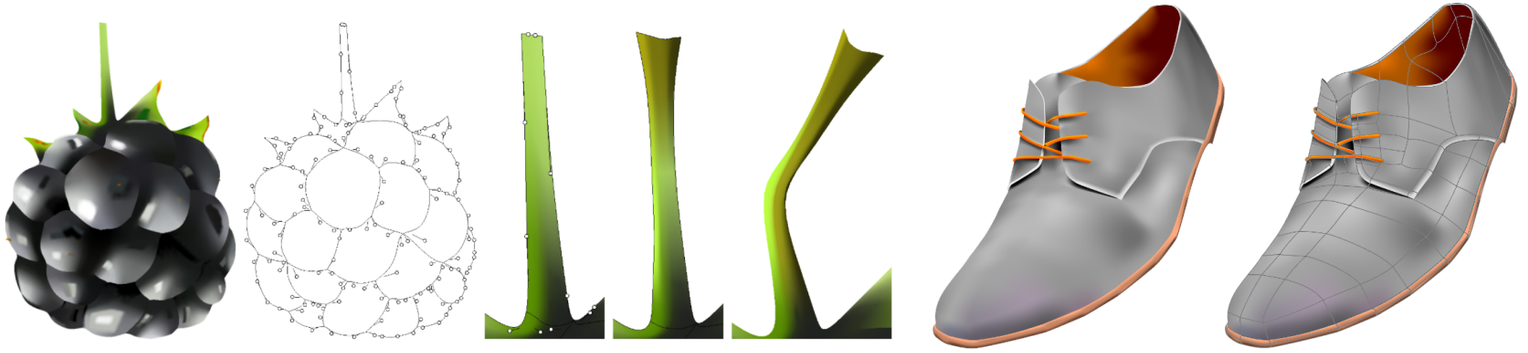
  - Alex Telea (Utrecht)



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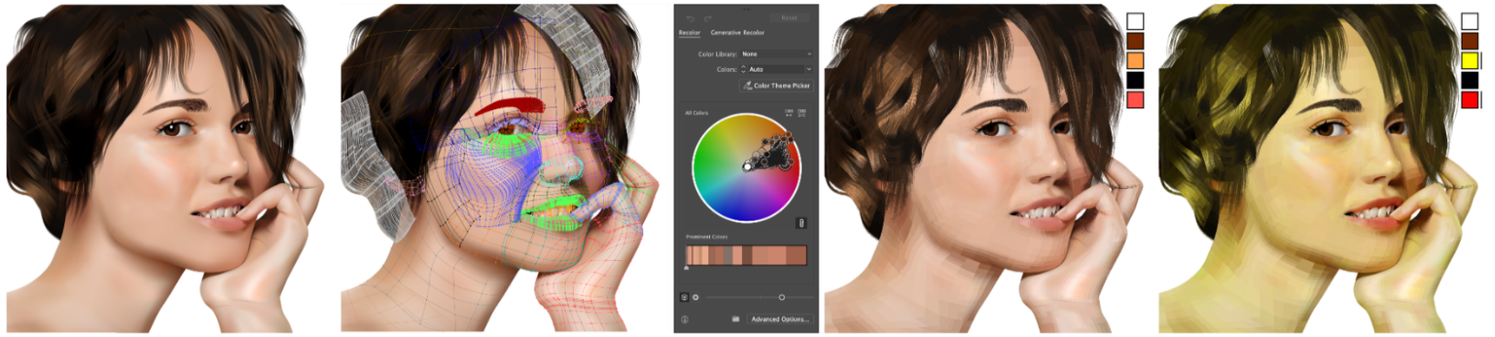
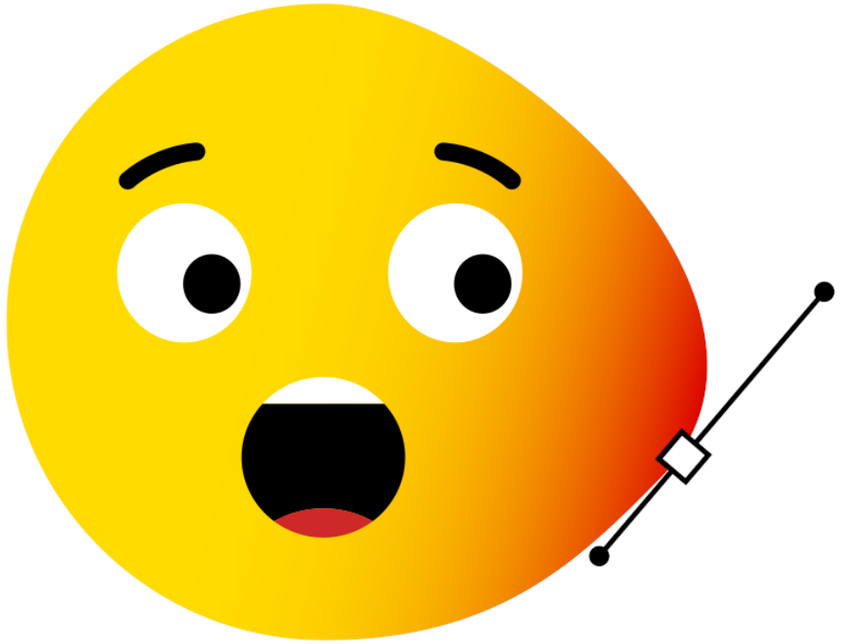
  - BSc: Sarah Baksteen, Rowan van Beckhoven, Rene Brals, Janeta Palii
  - MSc: Willard V. de la Houssaije, Jonathan Hogervorst, Eva Kato, Martijn Luinstra, Teun Verstraaten
  - PhD: Pieter Barendrecht, Kuanhao He, Gerben Hettinga, Henrik Lieng, Jingjing Shen, Flora Tasse, Jieying Wang, Jun Zhou

- Adobe Research
  - Jose Echevarria



Scientific Visualization and Computer Graphics



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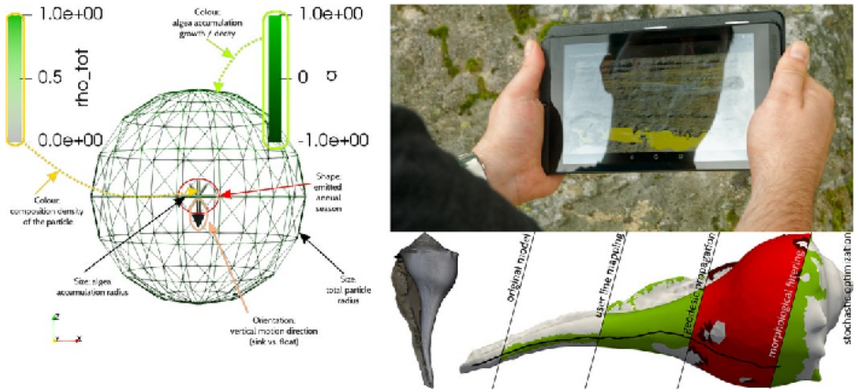
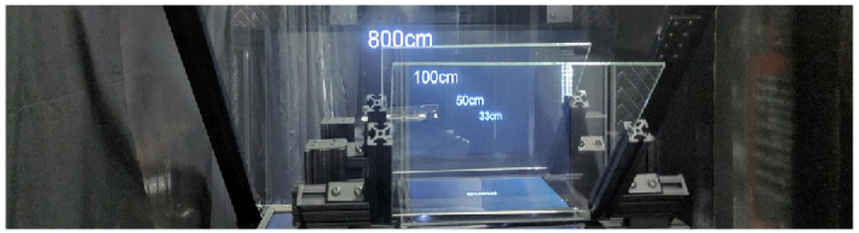
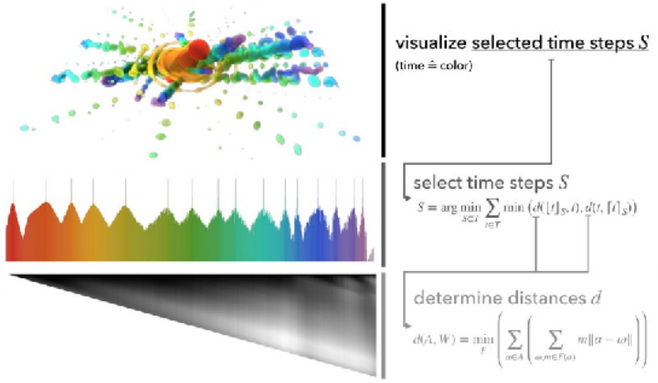
Scientific Visualization and Computer Graphics



Visualization- and Interaction Design



Perceptually optimized display systems



Optimization of scientific large data visualization

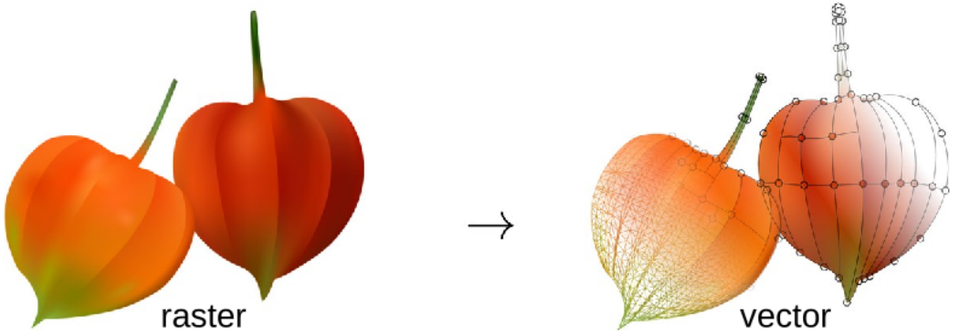


Image vectorization (fully scalable, sparse, easy to edit)

General topics and interests

- Shape and image modelling and processing
- Big data and high-performance visualization
- Optimization and machine learning
- Visual perception
- VR/AR display technology, eye tracking
- Data-driven method & system configuration
- Computer aided design and analysis
- Medical data, astronomy, …

Steffen Frey

Jiří Kosinka

Jos Roerdink

Alex Telea